SYSTEM ARCHITECTURE REFERENCE

Comprehensive System Architecture & Process Map

Canvas Marketing OS — reverse-engineered from source

A complete visual and written account of how the platform works end to end: what exists, why it exists, how it connects, what triggers it, how data moves, how processes execute, how agents interact, what depends on what, and where the architecture could be improved. Every architectural claim is traceable to a file in the repository.

40	19	10	14	4
DIAGRAMS	SECTIONS	SUBSYSTEMS	WIRED AGENTS	DB SCHEMAS

Repository rooipiet-tech/Canvas-Marketing-OS **Commit** 9956010 **Prepared** 17 August 2026 · rev 2

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System: Canvas Marketing OS (CMOS) **Repository:** `rooipiet-tech/canvas-marketing-os` **Document date:** 17 August 2026 · **Revision 2**, re-verified against `main` @ `e17a157` **Method:** Reverse-engineered from source. Every architectural claim below cites the file it was read from. Claims that could not be resolved from the repository are labelled **Unknown / Requires Confirmation**.

> **Revision 2 note.** Revision 1 was cut at `5c8ee07`. `main` has since advanced **59 commits / 138 files / ~17k lines**, and a large share of that work closes findings this document raised. Every fact below has been re-verified against the current tree — including F6, F9 and F11, which are now confirmed rather than carried forward. §14.0 summarises what moved.

How to read this document

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Diagram conventions. Used consistently in every diagram in this document:

SHAPE / STYLE	MEANING
`([Rounded])`	Human actor
`[Rectangle]`	Service or application
`[[Subroutine]]`	Agent (LLM-invoking function)
`[(Cylinder)]`	Database / schema / table
`[/Parallelogram/]`	Queue or message channel
`{Diamond}`	Decision point
`((Circle))`	Trigger / scheduled event
`[\Trapezoid\]`	External third-party system
Solid arrow `-->`	Synchronous call or control flow
Dashed arrow `-.->`	Asynchronous / event / best-effort

SECTION 1

Executive System Overview

1.1 What the system does

Canvas Marketing OS is an **agent-native marketing operations platform** built for Canvas Intelligence, a South African data-engineering firm selling to the office of the CFO. It automates the pipeline that a marketing team would otherwise run by hand:

> **scan the market → score what matters → write a brief → draft content → review it against brand and factual rules → get a human to approve it → schedule or publish it → measure what happened → report on cost and performance.**

It is not a content generator with a UI bolted on. It is a **governed workflow engine**: the interesting engineering is not in producing text, it is in the controls that sit around producing text — cost budgets, a data-redaction firewall, a client-naming permission register, an autonomy policy, cryptographic publish authorisation, kill switches, and an append-only audit trail.

1.2 Primary objectives

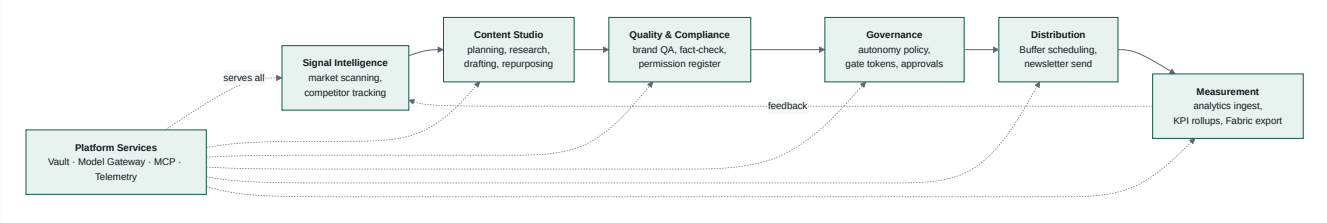
Five objectives are visible as first-class, enforced mechanisms in the code — each one has dedicated modules, database tables, and tests:

1. **Produce marketing output continuously and unattended.** Three declarative loops (`services/orchestrator/loops/*.yaml`) fire on schedules and fan out into task graphs.
2. **Never publish anything a human didn't authorise.** Autonomy levels (`services/gatekeeper/policy/autonomy.yaml`) + short-lived signed JWTs (`contracts/gate-token/spec.md`) + verification at the publish boundary (`services/publisher/app/verifier.py`).
3. **Never leak personal or client data.** A redaction firewall in front of every LLM call (`services/model-gateway/redaction.py`), metadata-only queue envelopes (`contracts/service-bus/spec.md`), a default-deny client-naming register (`functions/02-brand-steward-qa/permission_check.py`), and POPIA-shaped consent gating (`services/vault/vault/consent.py`).
4. **Keep spend bounded and attributable.** Every model call is metered to a `costs` row keyed by `agent_run_id` , with per-function daily budgets that downgrade the model tier before they block (`services/model-gateway/budget.py` , `metering.py`).
5. **Be operable by an agent, not only a human.** Every service exposes plain HTTP that a script can drive — refusal reasons included. This is stated explicitly as "agent-native by construction (AC-17)" in `services/publisher/main.py` and `services/orchestrator/main.py` .

1.3 Who and what interacts with it

ACTOR	NATURE	HOW IT INTERACTS
Marketing operator / approver ("Pieter" in code comments)	Human	Clicks Approve/Reject on a Teams Adaptive Card or an Entra-protected deep link; reads the Console
Console operator	Human	Browses tasks, traces, costs, Vault objects; flips the kill switch
Scheduled triggers	Machine	3 Logic Apps + 2+ Container Apps Jobs fire the loops
AI agents	Machine	14 wired LLM-invoking functions produce and review content
External SaaS	Machine	Anthropic, Buffer, Canva, GA4, Search Console, LinkedIn, Microsoft Teams, Mailchimp *(unwired)*, Microsoft Fabric
Automation agents / scripts	Machine	Drive every service over HTTP; e2e suite in `tests/e2e/` does exactly this

1.4 Major functional areas

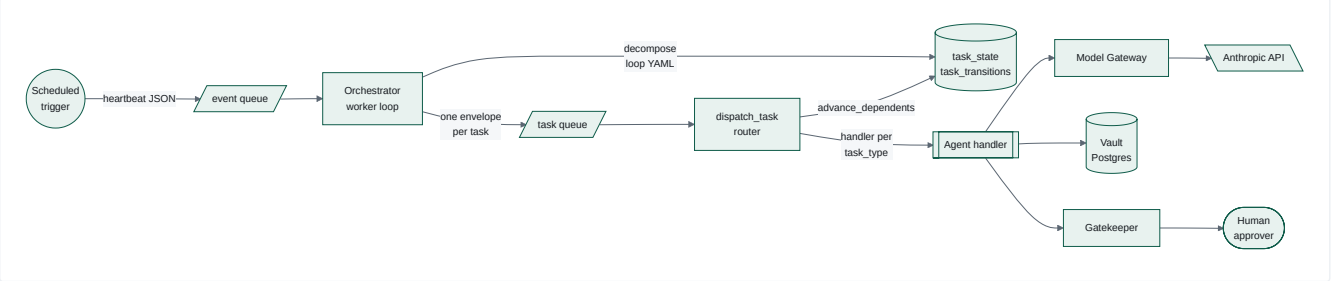


1.5 Major architectural components

COMPONENT	TYPE	RUNTIME	SOURCE
Orchestrator	Task-graph engine + FastAPI	Container App `ca-orchestrator`	`services/orchestrator/`
Shared policy	Scoring, scan profiles, source candidates as reviewed YAML	Read at runtime by dispatch handlers	`functions/_shared/`
Model Gateway	LLM broker with policy pipeline	Container App `ca-model-gateway`	`services/model-gateway/`
Gatekeeper	Autonomy policy + token issuer	Container Apps `ca-gatekeeper` (internal) + `ca-gatekeeper-approval` (external)	`services/gatekeeper/`
Publisher	Gate-token verifier + publish executor	Container App `ca-publisher` (internal)	`services/publisher/`
Vault	Domain object store API	Container App `ca-vault`	`services/vault/`
Analytics Ingest	Nightly ELT pipeline	Container Apps Job `caj-analytics-nightly-ingest`	`services/analytics-ingest/`
MCP Servers	Tool surfaces for agents	Container Apps `mcp-web`, `mcp-buffer`, `mcp-canva`	`mcp/`
Console	Operator web UI	Container App behind Easy Auth	`console/`
Registry	Function-package validator/signer	CI-only tooling	`services/registry/`
Telemetry Lib	Shared OTel wrapper	Python package, embedded	`services/telemetry-lib/`
Function packages	25 agent definitions	Prompts + schemas, read at runtime	`functions/`
Contracts	9 hash-frozen interface specs	Validated in CI	`contracts/`

1.6 How information and work flow through the system

The system has one dominant flow shape, repeated for every loop:



Three properties define this flow:

1. **The queue carries pointers, never payloads.** A `TaskEnvelope` holds only ids — `task_id`, `agent_run_id`, `campaign_id` — plus a string-only `metadata` bag. All real content is fetched from Postgres by id. This is a deliberate compensating control for running Service Bus Standard SKU without a private endpoint (`contracts/service-bus/spec.md`, `docs/accepted-risks.md`).
2. **The dependency graph is the control mechanism, not handler logic.** `db.advance_dependents()` flips a task `pending → dispatchable` only when *every* `depends_on` entry has reached `completed`. Isolation between sibling drafts is achieved by graph shape, by filtering inside a handler — this was an explicit redesign after a live incident (`services/orchestrator/loops/weekly-content-loop.yaml`, "ROUND 34" header comment).
3. **All tasks are published to the queue up front, at decompose time.** Nothing re-publishes them later. A task whose turn hasn't come raises `TaskNotReadyError` and its message is bounced back onto the queue, bounded at 20 requeues (`orchestrator/worker.py:NOT_READY_MAX_REQUEUES`). This is the single most consequential design decision in the system — see §14.

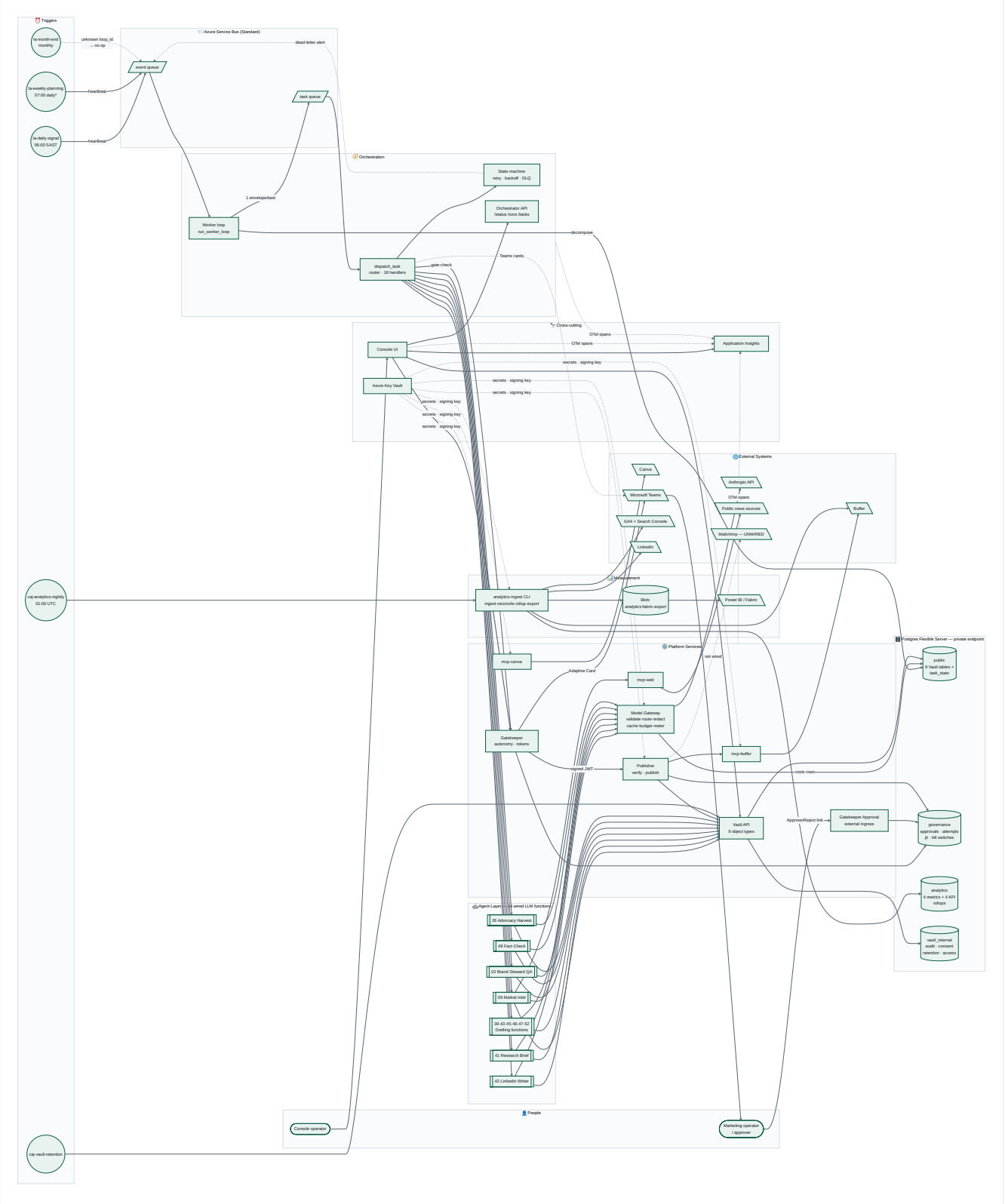
1.7 The most important design principles and architectural decisions

#	PRINCIPLE	EVIDENCE	CONSEQUENCE
D1	**Frozen contracts.** 9 interface files under `contracts/` are hash-guarded; breaking changes must land under `/v2/`.	`contracts/.frozen-v1.sha256`, `scripts/validate_contracts.py`	Extensions are made *additively* — e.g. `function_id` / `content_hash` are packed into the gate token's existing `resource` claim as canonical JSON rather than added as claims.
D2	**Policy is data, not code.** Routing, budgets, autonomy levels, fetch sources, loop graphs, redaction rules are all YAML.	`policy/routing.yaml`, `policy/budgets.yaml`, `policy/autonomy.yaml`, `loops/*.yaml`, `fetch_sources.yaml`	A model upgrade or an autonomy change is a reviewed one-line diff, not a deploy of new logic.
D3	**Fail closed.** Unlisted autonomy pair → level 0 (blocked). Unlisted client name → blocked identically to explicit UNCLEARED. Failed Vault lookup on publish → refuse.	`autonomy.yaml:default_level: 0`, `permission_check.py:check_clearance`, `publisher/app/vault_lookup.py`	Absence is never permission.
D4	**Append-only audit.** `gate_decisions` has no `updated_at` *by design*; `publish_attempts`, `approval_actions`, `task_transitions` are all insert-only.	`contracts/vault-schema/schema.sql` comments	Every branch — including every refusal — leaves exactly one row.
D5	**Defence in depth over single gates.** The kill switch is duplicated in Gatekeeper *and* Publisher with a parity test. Publish checks the content hash by recomputation *and* cross-checks the Vault. Redaction runs even though schema validation already ran.	`test_kill_switch_parity.py`, `routers/publish.py` ordering docstring	Costs duplication; buys independence.
D6	**Degrade, don't crash.** Missing `DATABASE_URL`, `SERVICE_BUS_NAMESPACE`, telemetry connection string, or Teams webhook are all normal states that log and continue.	`orchestrator/main.py` lifespan, `orchestrator/config.py`	Startup never fails on config absence — but see §14 on the observability cost.
D7	**Never guess a hostname.** Service URLs resolve via `az containerapp show` at runtime, or an env override. Never a hardcoded FQDN.	`orchestrator/clients/azure_fqdn.py`	Robust to redeploys; introduces an Azure-CLI runtime dependency (§14).
D8	**Errors never echo caller input.** Validation and routing messages are built from *schema-side facts only*, because they run before the redaction firewall.	`model-gateway/completion.py:validate_request`, `routing.py:unknown_model_message`	Closes an unscanned, unaudited exfiltration path.
D9	**Two independent state machines.** The Service Bus transport's `maxDeliveryCount` is fully decoupled from the application's own `retry_count`/backoff/dead-letter.	`orchestrator/state_machine.py` module docstring	Retries are observable in Postgres, not hidden in broker internals.

SECTION 2

Master System Architecture Diagram — Level 1

One picture of the entire platform. Everything below drills into a region of this map.

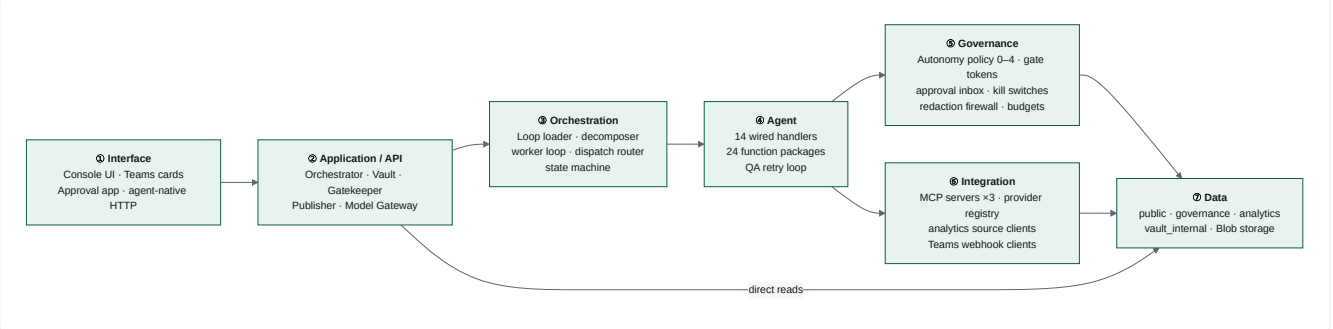


* The weekly trigger fires **daily** at 07:00 SAST. That is the deliberate cadence, not an override — one complete content cycle every morning for review. See §14.F3.

SECTION 3

Architecture Layers

The system resolves into **seven layers**. The layering is real, not aspirational: each boundary is enforced by network ingress rules (`external: false` on all but two apps), by contract validation in CI, or by both.

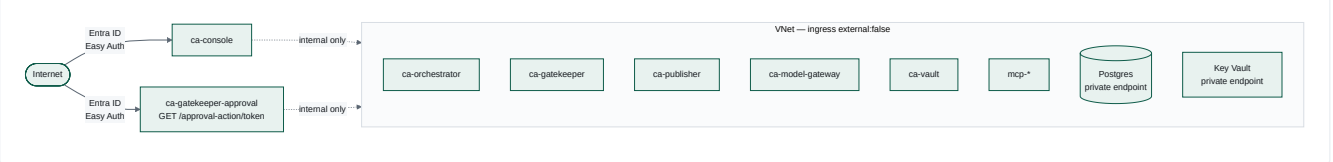


Layer responsibilities

LAYER	OWNS	MUST NOT	KEY EVIDENCE
① Interface	Human identity, rendering, deep links	Hold business logic; decide approvals from link possession	`console/app/auth.py` , `gatekeeper/app/routers/approval_action.py`
② Application/API	HTTP contract enforcement, error taxonomy	Bypass the layers below	`vault/main.py` exception handler, `model-gateway/main.py`
③ Orchestration	Task lifecycle, dependency resolution, retries	Know what a task *means*	`orchestrator/worker.py` , `state_machine.py`
④ Agent	Prompt assembly, output parsing, artefact creation	Call a provider SDK directly	`orchestrator/dispatch.py` , `functions/*/prompt.md`
⑤ Governance	Every allow/deny ruling and its audit row	Cache a decision	`gatekeeper/app/kill_switch.py` "NO CACHING, EVER"
⑥ Integration	Vendor protocol translation, credential use	Expose a publish path it wasn't given	`mcp/mcp-buffer/app/dispatch.py`
⑦ Data	Durable truth, FK integrity, retention	—	`contracts/vault-schema/schema.sql`

The two trust boundaries

Only **two** endpoints are reachable from outside the VNet:



Everything inside the VNet trusts its callers. `orchestrator/main.py` states this explicitly for `/tasks/{task_id}/review` : *"Internal-ingress-only, same trust boundary as /health, /status … no additional auth check, consistent with every other route here."* This is a deliberate, documented posture — and a real risk if the boundary is ever weakened (§14.S2).

SECTION 4

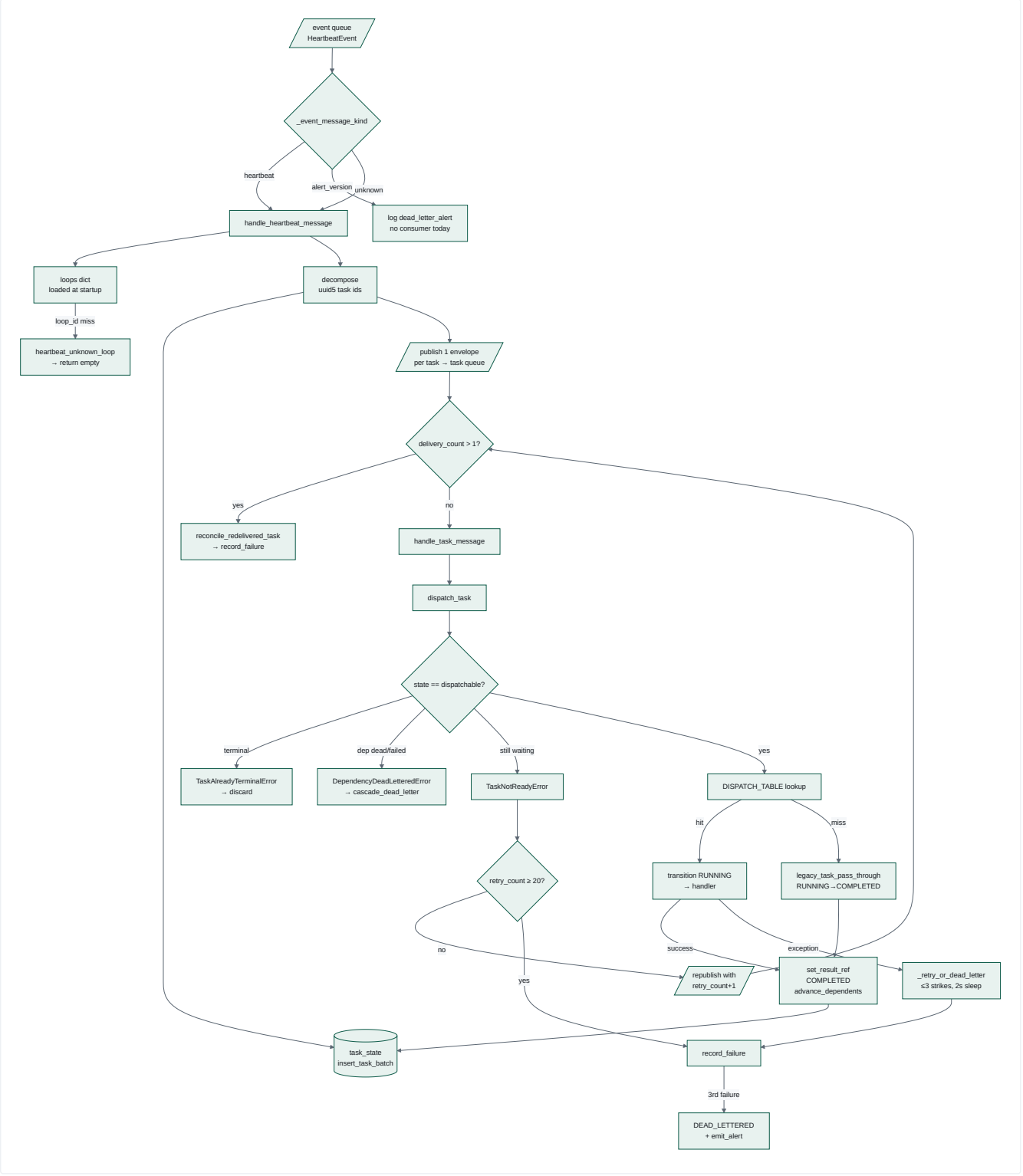
Subsystem Maps — Level 2

Ten subsystems. Each gets a fact table and its own node-and-edge diagram.

4.1 Orchestration & Task Lifecycle

Purpose. Turn a scheduled heartbeat into a persisted, dependency-ordered task graph, then drive every task to a terminal state exactly once.

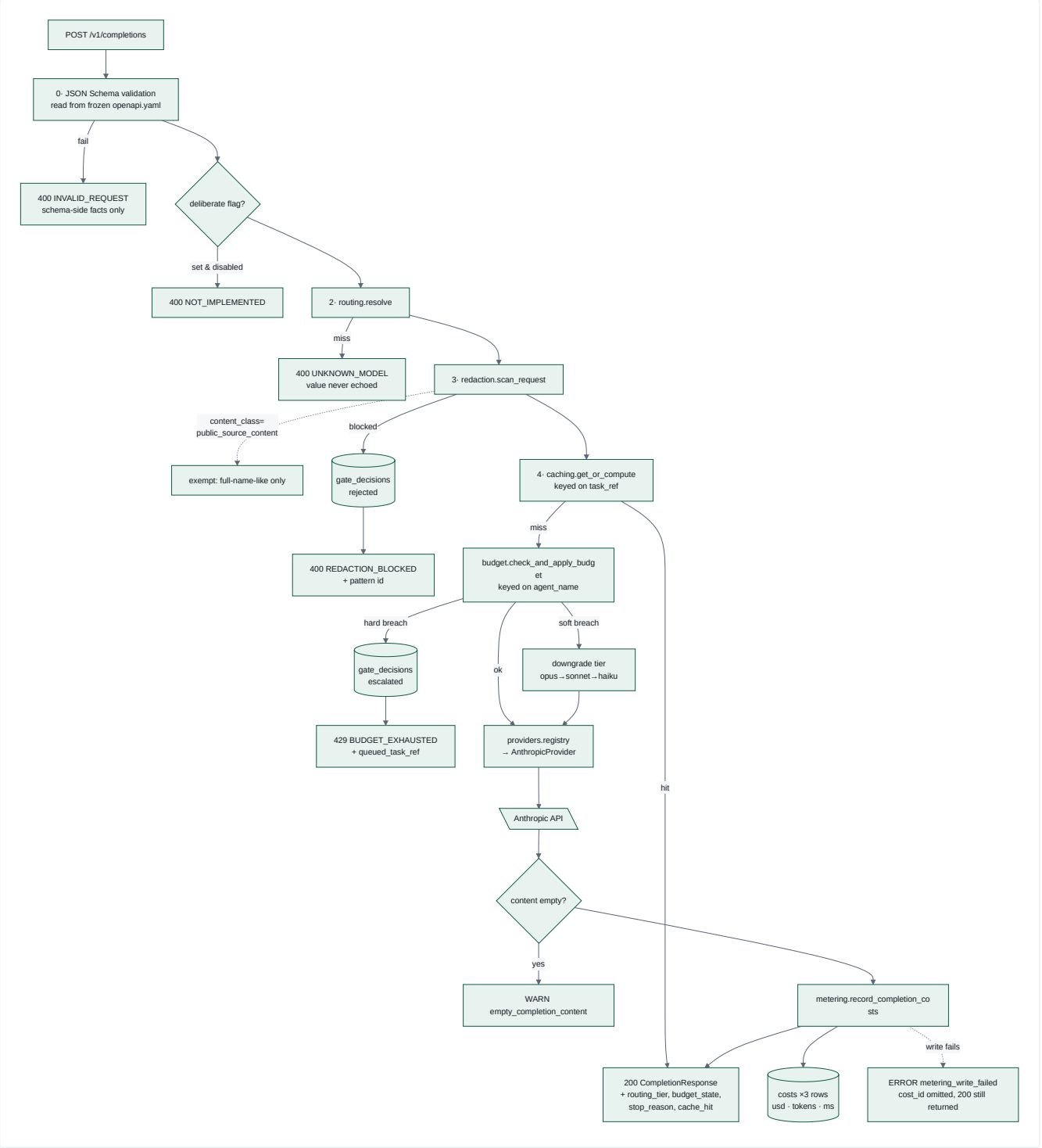
FACET	DETAIL
Components	`main.py` (FastAPI + lifespan), `worker.py` (poll loop), `decompose.py`, `loop_loader.py`, `dispatch.py` (router), `state_machine.py`, `dead_letter.py`, `db.py`, `run_state.py`, `task_review.py`, `servicebus/{producer,consumer,local_double}.py`
Inputs	`HeartbeatEvent` on `event` queue; `TaskEnvelope` on `task` queue; loop YAML at startup
Outputs	`task_state` / `task_transitions` rows; `TaskEnvelope` messages; `DeadLetterAlert` on `event`; `result_ref` JSONB payloads; HTTP status JSON
Dependencies	Postgres (`public.task_state`), Service Bus, Model Gateway, Vault, Gatekeeper, mcp-web, `functions/` prompt files, `contracts/` schema files
Upstream	Logic Apps triggers
Downstream	Every agent handler; Gatekeeper; Teams; Console
Data used	`task_state`, `task_transitions` (owned); Vault tables (via HTTP)
APIs exposed	`GET /health`, `GET /status`, `GET /runs/{task_ref}`, `GET /tasks/{task_id}/review`
Agents involved	All 14 wired agents, invoked through `DISPATCH_TABLE`
Triggers	Heartbeat messages; queue redelivery; internal requeue
Business rules	① A task dispatches only in state `dispatchable`. ② `advance_dependents` requires **all** deps `completed`. ③ 3 failures → `dead_lettered`. ④ A dep in `dead_lettered` / `failed` → immediate cascade, no retry. ⑤ Duplicate message on a terminal task → discard silently. ⑥ `task_id = uuid5(event_id, loop_task_id)` — deterministic and idempotent per heartbeat.
Failure points	Queue-bounce starvation (§14.C1); non-idempotent handler retries (§14.R1); worker loop is a single `asyncio.Task` inside the API process; `result_ref` lineage walk capped at `MAX_LINEAGE_HOPS = 6`



4.2 Model Gateway

Purpose. The single, provider-agnostic doorway to every LLM call, so cost, redaction, budget and provider swaps happen in exactly one place.

FACET	DETAIL
Components	`main.py` (router + exception mapping), `completion.py` (the pipeline), `routing.py`, `redaction.py`, `budget.py`, `caching.py`, `metering.py`, `gate_decisions.py`, `providers/{base,registry,anthropic}.py`, `db.py`
Inputs	`CompletionRequest` JSON: `model`, `messages`, `agent_run_id`, optional `max_tokens`, `temperature`, `tools`, plus additive `task_ref`, `deliberate`, `content_class`
Outputs	`CompletionResponse` with `content`, `usage`, `cost_id`; 3 `costs` rows; `gate_decisions` row on block/breach; one JSON log line per request
Dependencies	`contracts/model-gateway/openapi.yaml` (read at runtime), `redaction-rules.yaml`, Anthropic API, Postgres
Upstream	Every agent handler in `dispatch.py`; registry eval harness
Downstream	Anthropic; `public.costs`; `public.gate_decisions`
Business rules	① Validate against the frozen schema before touching any field. ② Never echo a submitted value in an error. ③ Redact before any provider call. ④ Same `task_ref` ⇒ one upstream call. ⑤ Soft breach (≥80% of daily limit) downgrades tier; hard breach returns 429 + `gate_decisions` escalation, provider never called. ⑥ Metering failure must not fail a paid-for completion.
Failure points	In-process cache is per-replica only (§14.B2); startup model-liveness check is advisory; price table is hand-maintained (§14.M1)



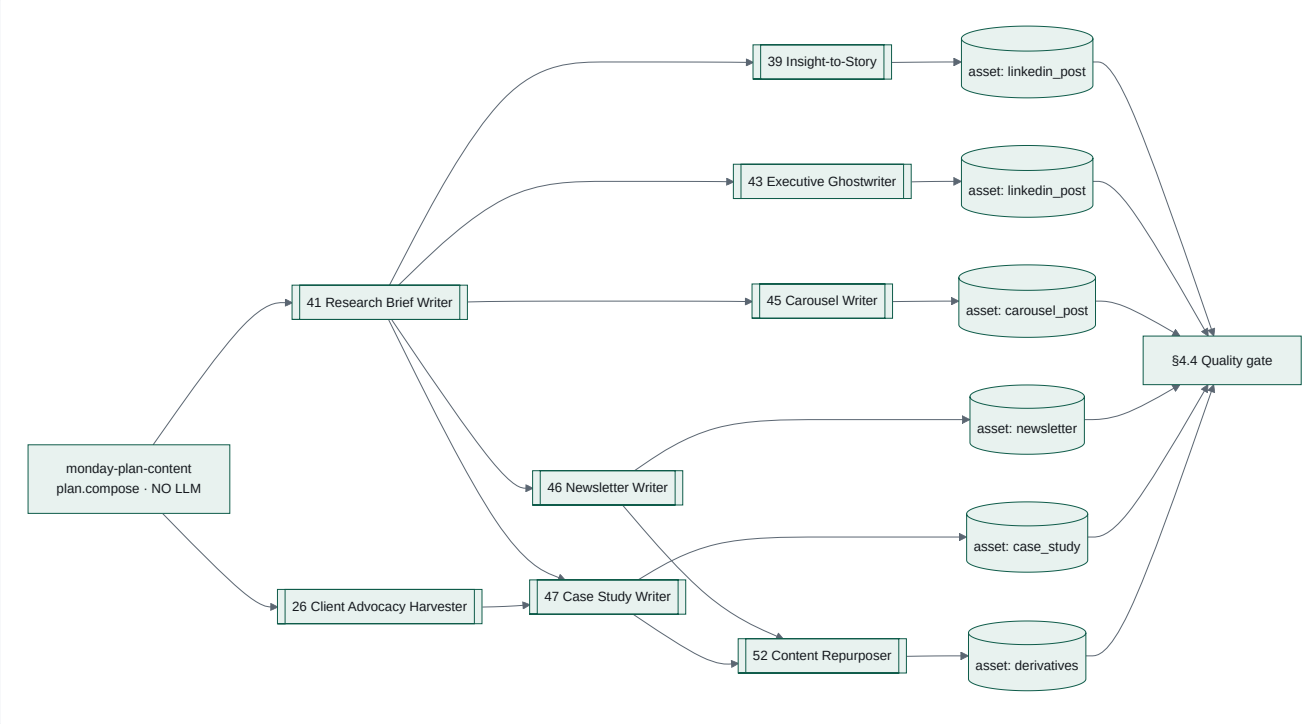
Redaction scope — stated precisely because it has been narrowed twice after live incidents:

SCANNED	NOT SCANNED	WHY
`messages[*].content` where role ∈ {user, assistant, tool}	role = `system`	System prompts are static, developer-authored files (`_read_prompt()`); the universal LLM convention makes them operator instructions, never end-user content
serialized `tools[]` payload	—	The one contract field with `additionalProperties: true` — a messages-only scanner would be bypassable
all 4 heuristic patterns + fixture exact-matches	`full-name-like` , **only** when `content_class == "public_source_content"`	Public news prose and positioning copy trip "two consecutive Title-Case words" constantly; 4 authorised call sites in `dispatch.py` only

4.3 Agent / Content Studio Subsystem

Purpose. Convert a plan and a research brief into six reviewable content assets per week, plus the daily brief chain.

FACET	DETAIL
Components	18 handlers in `dispatch.py` ; 24 function packages in `functions/` ; shared helpers `_draft_social_post_handler` , `_render_*` , `_parse_json_content` , `resolve_lineage_result`
Inputs	`TaskEnvelope` ; ancestor `result_ref` ; Vault briefs/assets; `prompt.md` files
Outputs	Vault `assets` (with `content_hash`), `briefs` , `signals` ; `agent_runs` rows; `result_ref`
Dependencies	Model Gateway, Vault, `FUNCTIONS_DIR` staged into the container image
Business rules	① Every function returns a single bare JSON object; a chatty model is tolerated by `raw_decode` + trailing-content logging. ② Client naming is default-deny. ③ Every claim needs a proof point. ④ Case-study drafts deliberately have **no** Friday Buffer task — human-initiated cadence only. ⑤ `max_tokens` is sized per asset type (1536 → 4096) after a live truncation incident.
Failure points	Prompt files must be staged into the image (`FUNCTIONS_DIR`) — a class of bug that has bitten twice (§14.P1); JSON parse failures are the most common dead-letter cause historically



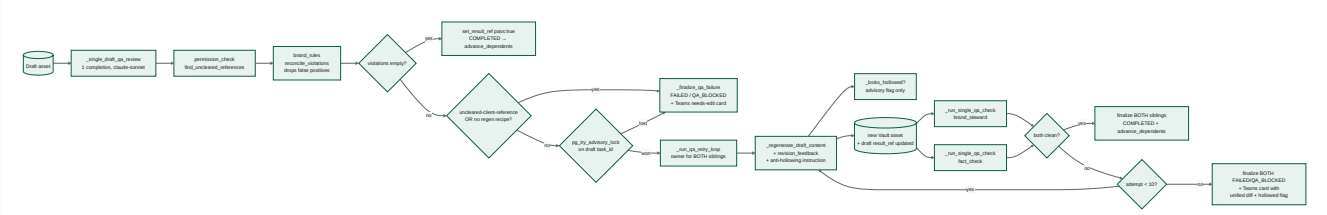
Function package anatomy — every package under `functions/NN-name/` carries the same five artefacts, validated by `services/registry/validate_package.py` :

FILE	ROLE
`prompt.md`	The system prompt sent to the gateway — the agent's actual instructions
`schema.json`	Input contract
`tools.yaml`	Declared tool surface + permission tier, validated against `contracts/function-definition/tools.schema.json`
`skill.md`	Human-facing purpose / when-to-invoke / when-NOT-to-invoke
`evals/task-0N-*.json`	5 golden eval tasks with rubrics, run in CI

4.4 Quality & Compliance Subsystem

Purpose. Stop non-compliant content before it can reach a gate-check — and, since 11 Aug 2026, try to *fix* it automatically first.

FACET	DETAIL
Components	`qa_review_handler` , `_single_draft_qa_review` , `_run_qa_retry_loop` , `_run_single_qa_check` , `_regenerate_draft_content` , `_looks_hollowed` , `_finalize_qa_failure` , `brand_rules.reconcile_violations` , `functions/02-brand-steward-qa/permission_check.py` , `services/registry/safety_suite.py`
Inputs	Draft asset bytes; `client_references` ; channel; `docs/permission-register.yaml`
Outputs	`pass` / `violations` verdict; regenerated assets; `FAILED` / `QA_BLOCKED` transitions; Teams "needs edit" / "retries exhausted" cards
Business rules	① `pass` is true only when `violations` is empty — no partial pass. ② `uncleared-client-reference` is **never retryable**. ③ Every retry re-runs *both* review kinds, not just the failing one. ④ Anti-hollowing detection is advisory only — it never blocks a retry. ⑤ A Postgres advisory lock on the draft's `task_id` makes exactly one sibling the retry owner; the loser finalises immediately rather than blocking.
Failure points	A losing sibling may emit a "needs edit" card that the winner supersedes seconds later — documented and accepted; `draft-content-repurpose` has no regeneration recipe and falls back to single-shot



The six Brand Steward checks (`functions/02-brand-steward-qa/prompt.md`):

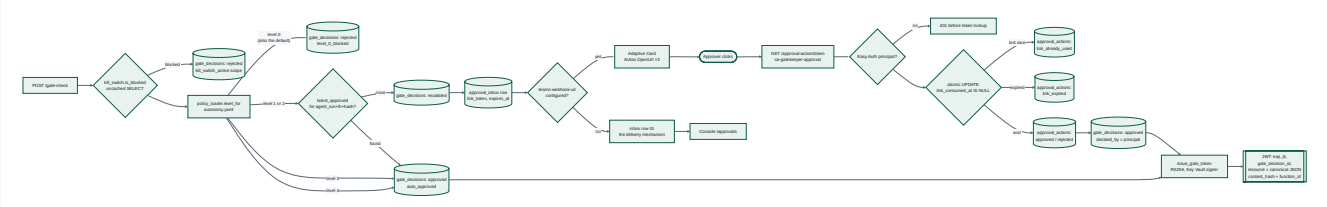
#	VIOLATION CODE	RULE	EXEMPTION
1	`uncleared-client-reference`	Default deny; absence ≠ permission	none — nothing in the register is CLEARED today
2	`link-shortener`	No bit.ly / lnkd.in / tinyurl / ow.ly / buff.ly	none
3	`sa-english-spelling`	SA English required	none
4	`missing-cta`	Exactly one CTA	`channel: internal-brief`
5	`url-utm`	Full canvasintelligence.com URL with utm_source/medium/campaign	`channel: internal-brief`
6	`unsupported-claim`	Proof over platitude — a superlative needs a number, artefact or CLEARED client	none

The same six codes are also implemented **deterministically** (regex + lexicon, no model) in `services/registry/safety_suite.py` for CI. Two independent implementations of the same policy — see §14.D1.

4.5 Governance Subsystem (Gatekeeper)

Purpose. Decide whether an action may happen, record that decision immutably, and — only if approved — mint a short-lived cryptographic authorisation.

FACET	DETAIL
Components	`main.py` (internal), `approval_main.py` (external), `routers/{gate_check,approval_action,approval_status,decisions}.py` , `policy_loader.py` , `approval_inbox.py` , `tokens.py` , `signer/{keyvault,local}_signer.py` , `kill_switch.py` , `teams_client.py` , `auth.py`
Inputs	`GateCheckRequest` : `agent_run_id` , `function_id` , `action_class` , `content_hash` , preview fields
Outputs	Exactly one `gate_decisions` row per call; optionally an `approval_inbox` row + Teams card + approve/reject URLs; on approval, an RS256 JWT
Business rules	① Kill switch checked **first**, uncached, every call. ② Unlisted `(function_id, action_class)` → level 0. ③ No `publish` entry may sit above level 2 (test-enforced). ④ No `smoke.*` / `test.*` function_id may exist (RISK-01). ⑤ Approval links are single-use via atomic conditional UPDATE and expire in 24h. ⑥ The approver is the Easy-Auth principal, never the link holder.
Failure points	Key Vault availability for signing; a level-2 "elevated" tier is documented as reserved but behaves identically to level 1 today



Autonomy levels (`services/gatekeeper/policy/autonomy.yaml`):

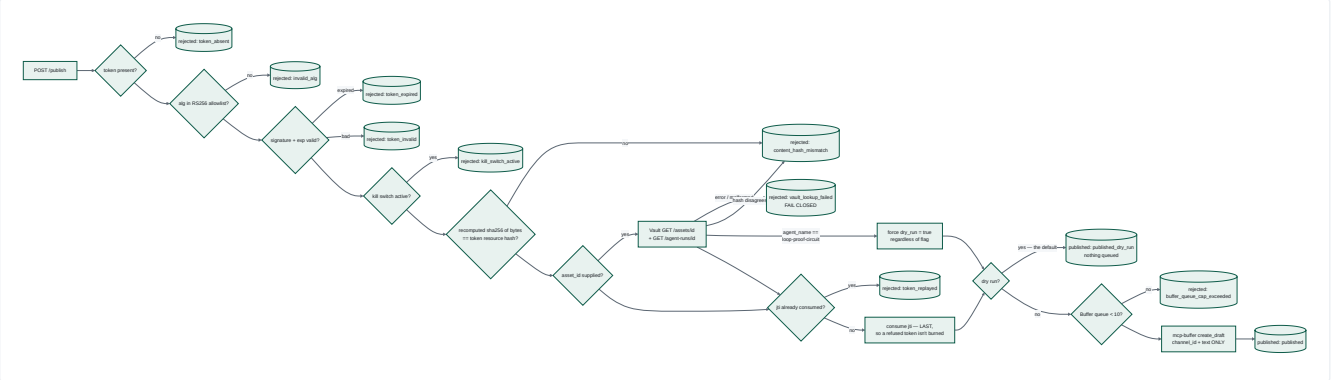
LEVEL	MEANING	SHIPPED ENTRIES
0	Blocked always — no approval can unblock	`publish.paid_ad` ; **and every unlisted pair**
1	Single human approver	`publish.social_post`
2	Elevated approval (same mechanism today; reserved for quorum)	`publish.blog_article`
3	Auto-approved and audited	`draft.social_post` , `draft.brief`
4	Fully autonomous, logged	`analyse.signal` , `analyse.campaign_performance`

4.6 Publishing & Distribution Subsystem

Purpose. Be the only thing that can turn an authorisation into an external side effect — and refuse, auditably, on any doubt.

FACET	DETAIL
Components	`routers/publish.py` , `routers/publish_attempts.py` , `verifier.py` , `jti_ledger.py` , `hashing.py` , `kill_switch.py` , `vault_lookup.py` , `buffer_client.py` , `esp_client.py` , `vault_adapter.py`
Inputs	`PublishRequest` : gate token, raw content bytes, `function_id` , `agent_run_id` , optional `asset_id`
Outputs	Exactly one `governance.publish_attempts` row per call; a Buffer draft in live mode; `jti_ledger` consumption
Business rules	The ordering is the design — see below
Failure points	`esp_client.py` is written but **not wired** into the router; `buffer_client.resolve_live_fqdn` shells out to `az` at runtime

Check ordering rationale (from `routers/publish.py` 's docstring — each position is deliberate):



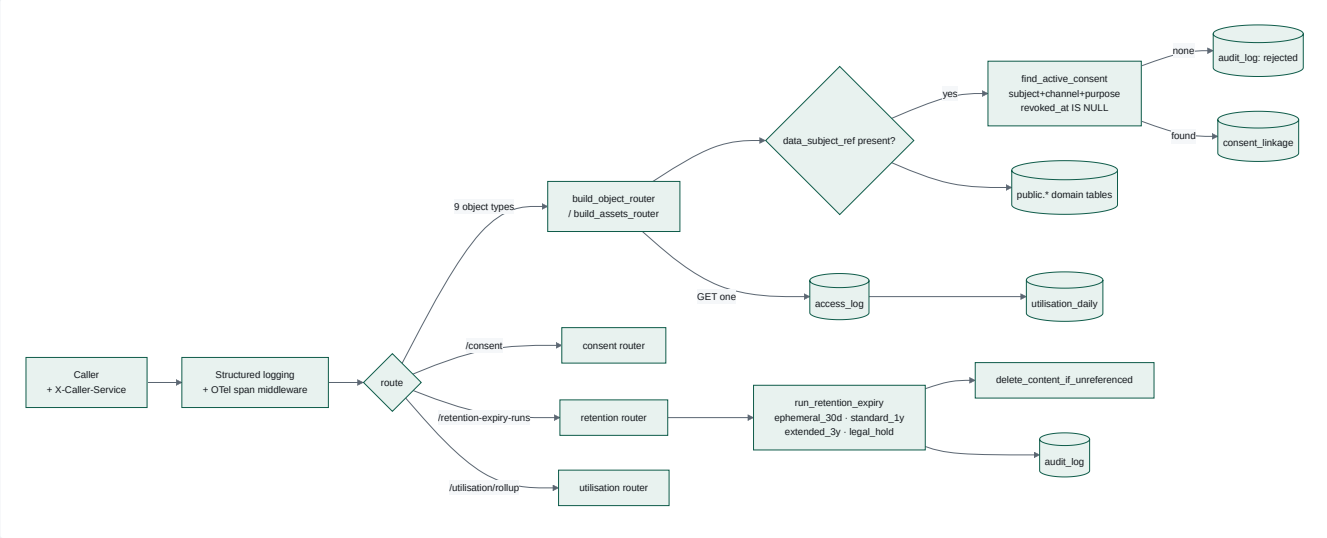
Three ordering facts worth stating plainly:

- **Kill switch is re-checked after token verification but before consumption** — a pre-issued, still-valid token cannot outlive an operator flipping the switch.
- **The Vault cross-check runs before the jti burn** — a broken lookup fails closed *without* spending the token.
- **The jti is consumed last** — a token refused for any other reason remains usable once the cause is fixed.

4.7 Vault (Data Custody) Subsystem

Purpose. Own the domain objects, and enforce POPIA-shaped consent, retention and access accounting on top of them.

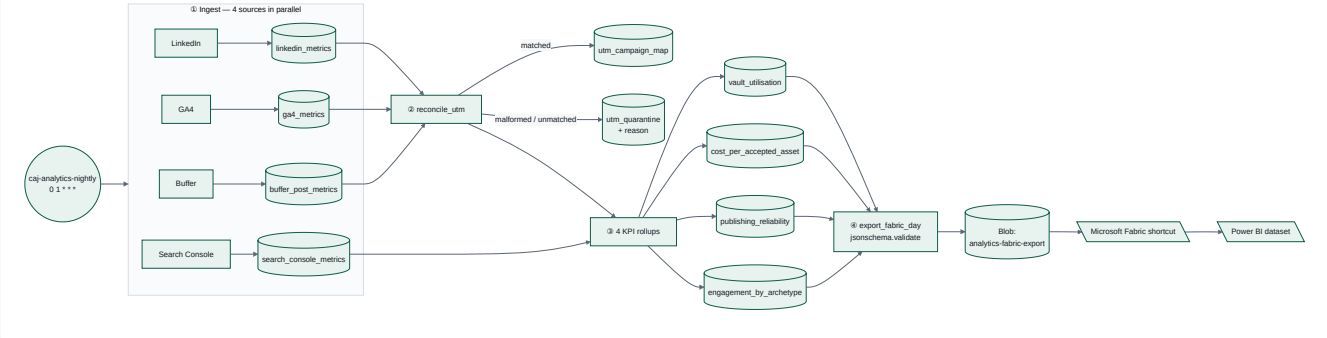
FACET	DETAIL
Components	`main.py` (+ structured request-logging middleware), `routers/{objects,consent,retention,utilisation}.py` , `models.py` (9-type registry), `consent.py` , `retention.py` , `storage.py` , `rollup.py` , `audit.py` , `db.py`
Inputs	HTTP CRUD on 9 object types; `X-Caller-Service` header
Outputs	Domain rows; `vault_internal.{access_log, utilisation_daily, consent_linkage, retention_policy, audit_log}` ; one JSON log line + `X-Correlation-Id` per request
Business rules	① A create carrying `data_subject_ref` is rejected unless a matching non-revoked `consent_register` row exists — and the accepted write is durably linked to it. ② Retention classes map to fixed durations; `legal_hold` uses a year-9999 sentinel so the NOT NULL constraint holds uniformly. ③ Every individual-object GET writes an `access_log` row, rolled up daily.
Failure points	Console's Vault search fetches full lists then filters client-side — explicitly flagged in-code as interim (§14.B3)



4.8 Analytics & Measurement Subsystem

Purpose. Close the loop — pull performance data back from every channel, reconcile it to campaigns, compute four KPIs, and hand it to Fabric/Power BI.

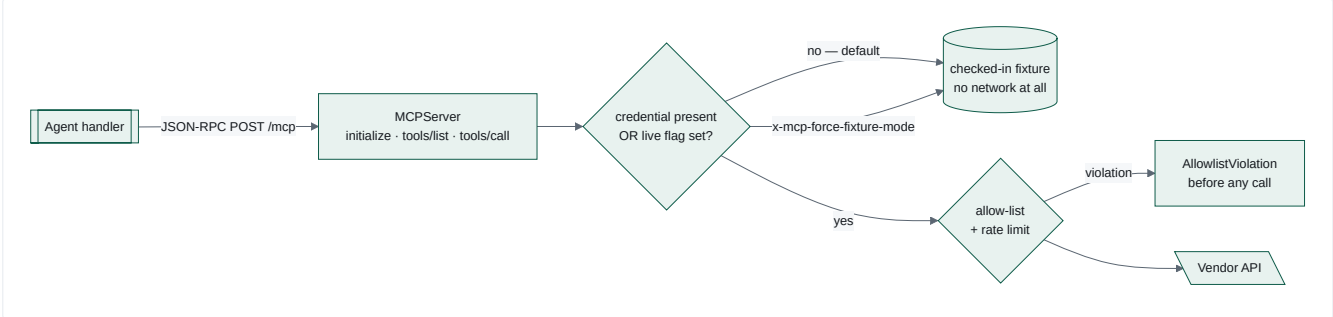
FACET	DETAIL
Components	`cli.py` (`run` / `nightly`), `ingest.py` , `utm.py` , `rollup.py` , `fabric_export.py` , `blob_writer.py` , source clients `{buffer,ga4,search_console,linkedin}_client.py` , `google_auth.py` , `credentials.py` , `vault_client.py`
Trigger	Container Apps Job `caj-analytics-nightly-ingest` , cron `0 1 * * *` (01:00 UTC / 03:00 SAST) running `python -m analytics_ingest.cli nightly --day <yesterday>`
Business rules	① Every rollup upserts `ON CONFLICT … DO UPDATE` — re-runs are idempotent. ② Rows whose `utm_campaign` doesn't match `utm_campaign_map` are quarantined with a reason, never silently dropped. ③ Groups with zero impressions / zero accepted assets are skipped rather than dividing by zero. ④ The export self-validates against `analytics/contracts/fabric-nightly-export.schema.json` before upload.
Failure points	Dual-mode fixtures vs. live — only Buffer has a live path in this session; the loop file documents a graph the orchestrator does **not** execute (§14.F2)



4.9 MCP Tool Subsystem

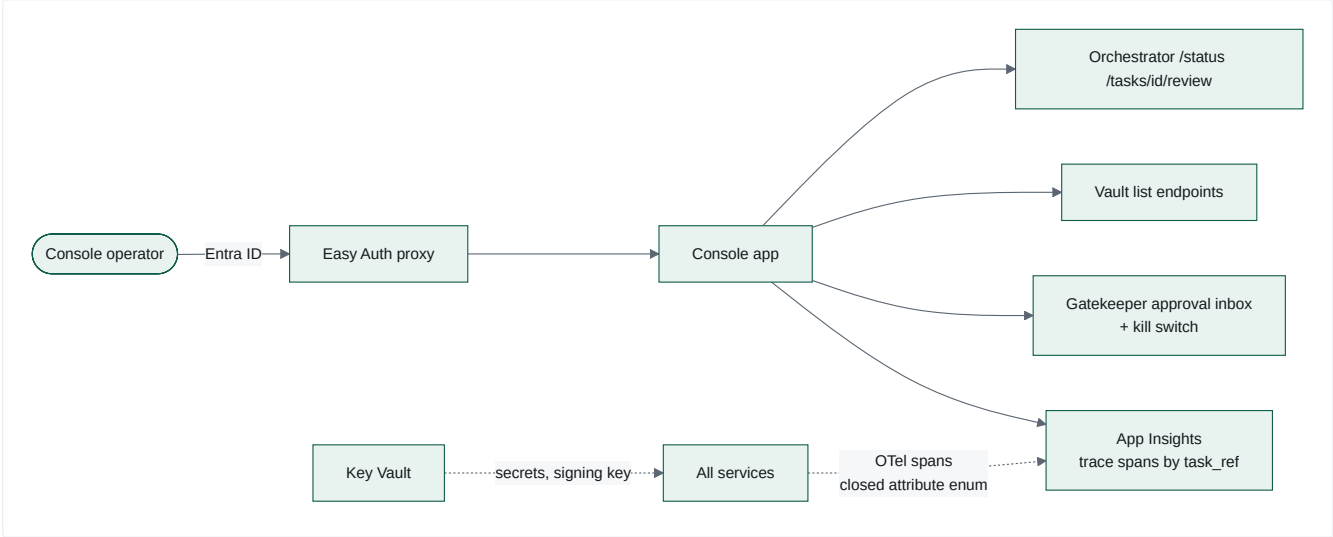
Purpose. Give agents a uniform, permission-scoped tool surface, and keep every vendor credential behind it.

SERVER	TOOLS	MODE GATE	NOTABLE CONSTRAINT
mcp-web	`fetch_url`	`MCP_WEB_LIVE_MODE` (non-secret flag, waivered — this server has no vendor credential)	Egress allow-list checked *before* any network call; sliding-window rate limit
mcp-buffer	`list_queue` , `get_post` , `create_draft`	Presence of `BUFFER_API_KEY` / Key Vault secret	`create_draft` status is **hardcoded server-side**; no tool exposes a status/mode/state parameter; no name or description may match `publish|share.?now|send.?now|go.?live` (pytest-enforced)
mcp-canva	`create_design_from_template` , `bulk_create_from_csv` , `export_design`	Credential presence	Both creation tools are **template-locked** — `template_id` is required, so no free-form design creation path exists



4.10 Console, Telemetry & Delivery Subsystem

FACET	DETAIL
Console	FastAPI + Jinja2, behind Container Apps Easy Auth with `unauthenticatedClientAction: Return401`. Read routes: `/tasks`, `/tasks/{ref}/trace`, `/review/{task_id}`, `/approvals`, `/vault-search`, `/costs`, `/kill-switch`. One write route: `POST /kill-switch/toggle`, same-origin-checked, operator identity taken from Easy Auth headers. `/health` is deliberately unauthenticated for probes.
Telemetry	`services/telemetry-lib` wraps OpenTelemetry with a **closed enum** of span attribute keys (`function_id`, `task_ref`, `model`, `registry_version`, `cost` required; `status`, `duration_ms`, `error_code` optional) and structurally rejects values over 200 chars — mirroring the queue's no-free-text rule for spans. Every service wires `traceparent` propagation. Exporter: Application Insights.
Registry / CI	`services/registry` validates package shape, builds a **byte-reproducible** canonical-JSON manifest signed with Ed25519 (no timestamps, no absolute paths, CRLF normalised), runs golden evals in mocked mode with a per-rubric assertion of zero live calls, and runs the deterministic safety suite. 15 GitHub Actions workflows cover CI, per-service image builds, infra deploy via OIDC federated identity (no client secrets), and Slack/Teams deploy notification.



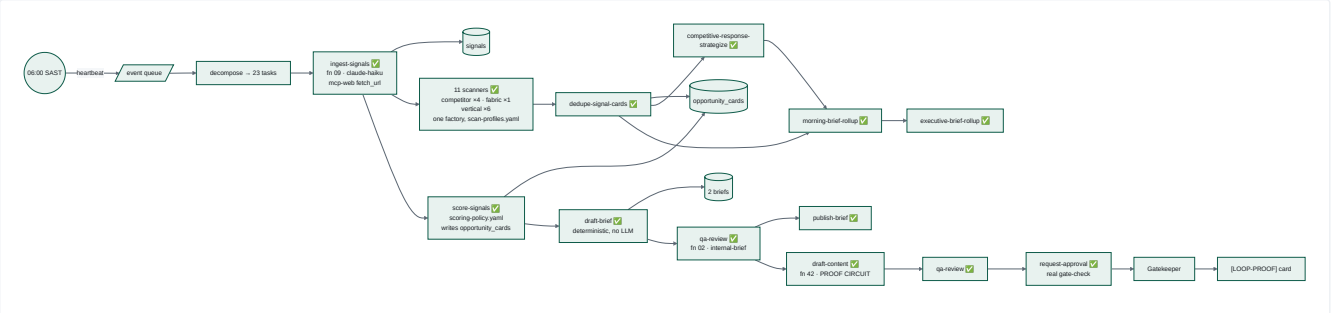
SECTION 5

End-to-End Process Flows — Level 3

Five processes matter. Each is traced **Trigger → Input → Processing → Decisions → Agents/Services → Data → External → Output → Next action**.

5.1 Process A — Daily Signal Loop

Trigger: Logic App `la-daily-signal-loop-trigger`, 06:00 South Africa Standard Time, daily. **Status (revision 2):** all **23 tasks now have real handlers**. At revision 1 only 6 did; the other 17 were `legacy_task_pass_through` no-ops. The eleven S10 scanners are registered from one factory (`SCANNER_TASKS` → `_make_scanner_handler` → `**SCANNER_HANDLERS`), so a naive grep of `DISPATCH_TABLE` still under-counts them — resolve the spread before concluding anything is unwired.



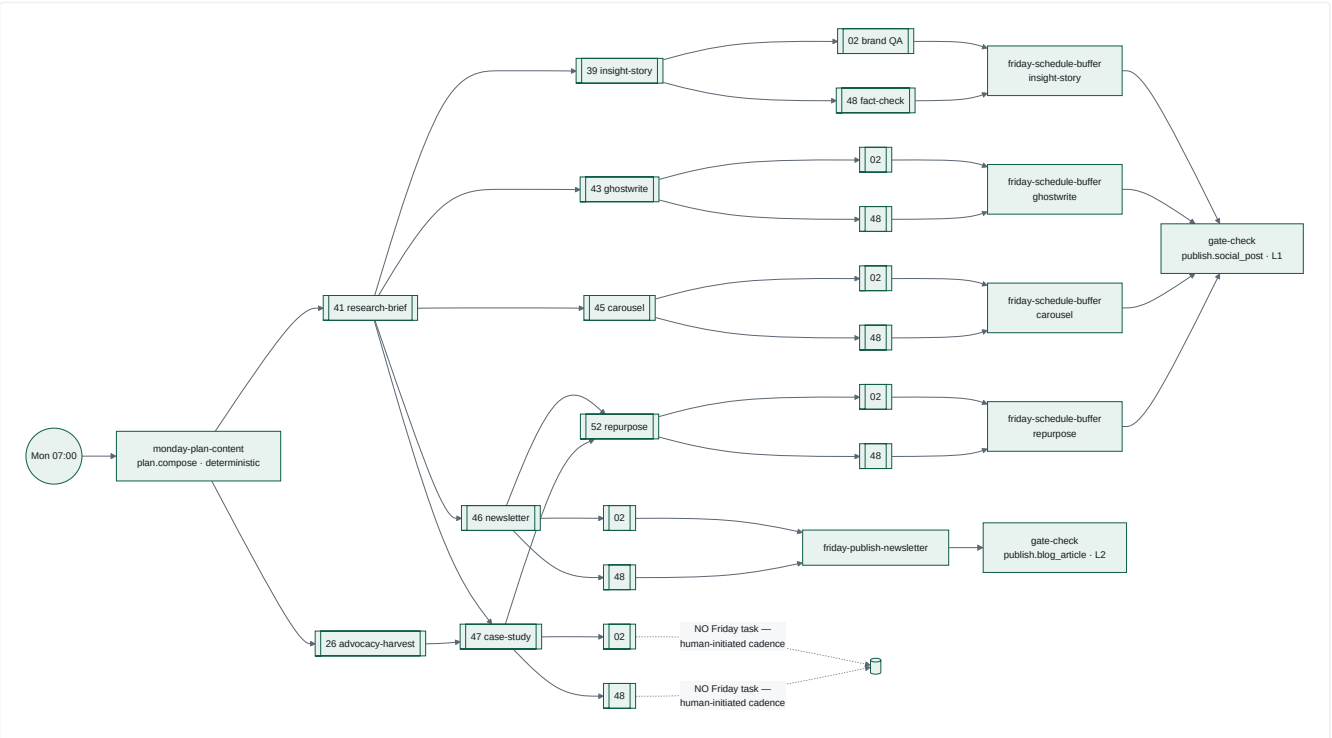
The S8 Proof Circuit deserves a note. It is a deliberately isolated, permanently dry-run exercise of the full `signal → brief → draft → QA → approval-card` path against the **live** platform. Its isolation is enforced in three independent places:

1. `params.proof_circuit: true` in the loop YAML → carried onto `TaskEnvelope.metadata` by `worker._task_metadata`.
2. Every Vault `agent_run` it creates is tagged `agent_name = "loop-proof-circuit"`.
3. **Publisher forces** `dry_run = true` whenever an asset's `agent_name` resolves to that constant — *regardless of the* `PUBLISHER_DRY_RUN` *flag* (`publisher/app/vault_lookup.py`). The constant is duplicated in both services and kept honest by a cross-service test.

5.2 Process B — Weekly Content Loop

Trigger: Logic App `la-weekly-planning-trigger`, **07:00 SAST every day**. Daily is the deliberate cadence (confirmed 17 Aug 2026) — one complete cycle lands each morning for review. **All 27 tasks have real handlers.** This is the production content pipeline.

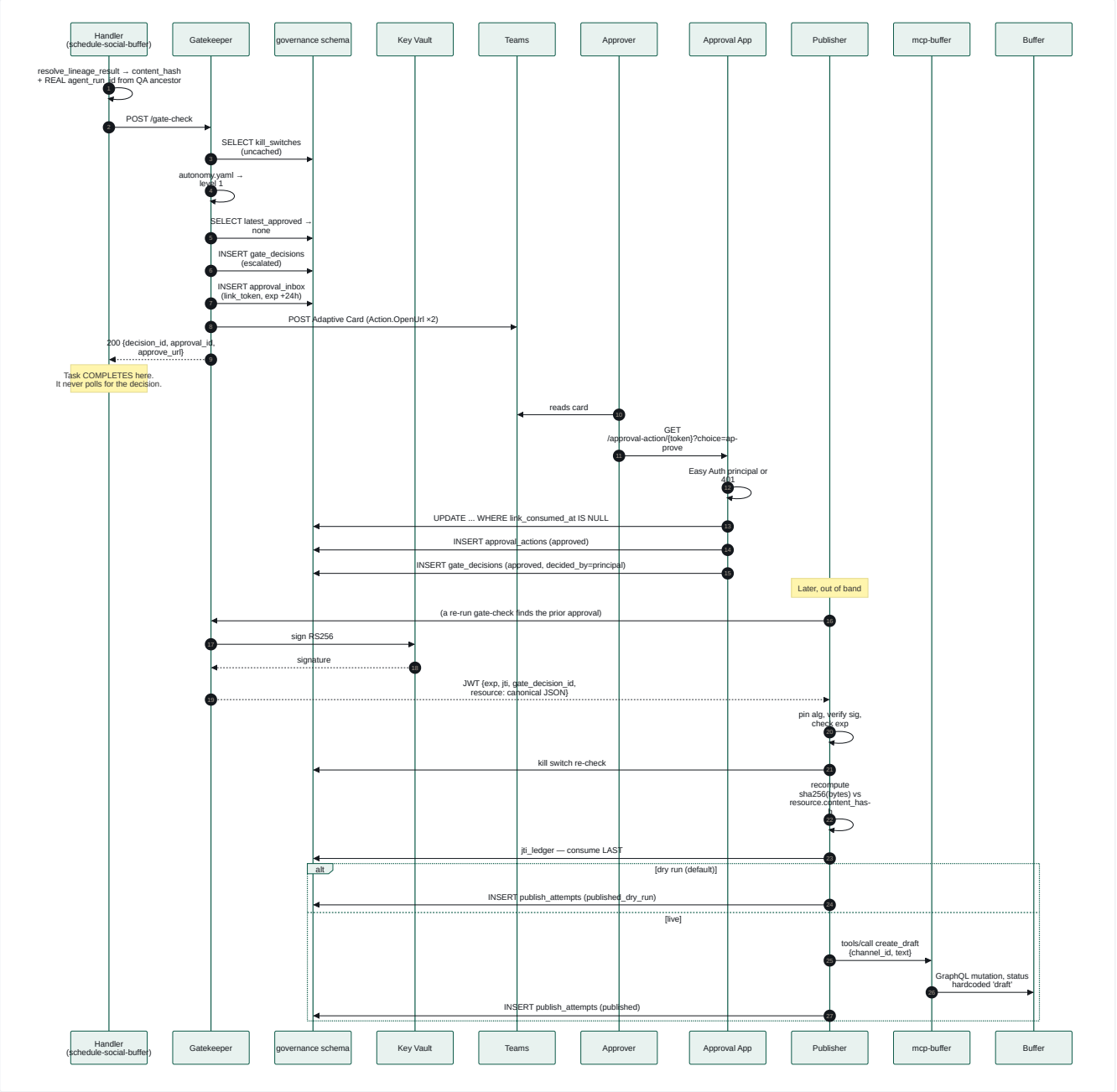
The loop id and its `monday-` … `friday-` task-id prefixes are **dependency-chain names, not a schedule.** One heartbeat decomposes the whole graph and runs it as fast as `depends_on` allows, so a daily fire means a full Mon-Fri cycle per day — not one weekday's slice per day.



Why the graph looks like this. It originally had **two** aggregate Thursday tasks, each depending on all six Wednesday drafts and resolving to one all-or-nothing state. On the night of 10 Aug 2026, a **spelling typo in one draft** dead-lettered the Friday publish tasks for every other draft — including 4–5 that individually passed both reviews cleanly. The fix was a full restructure to 12 per-draft review tasks. The isolation comes from the dependency graph shape, not from handler-level filtering. This is the clearest example in the codebase of a load-bearing architectural principle: *let the graph enforce isolation.*

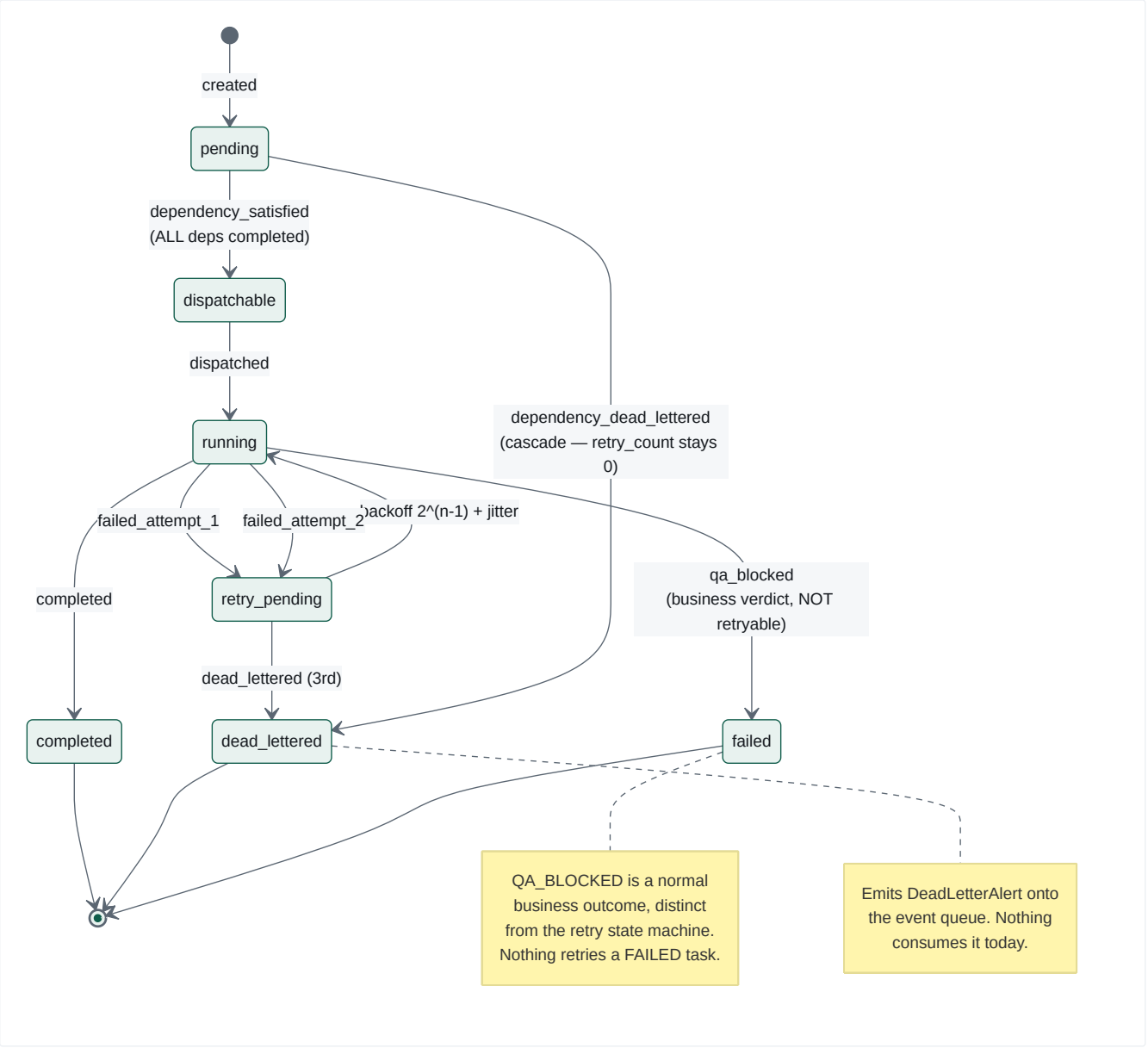
5.3 Process C — Human Approval & Publish

The most security-sensitive path in the system.



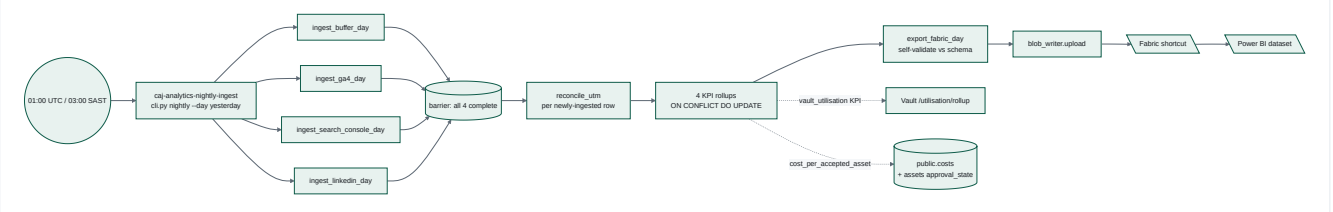
Note the gap. The handler completes as soon as `/gate-check` responds. Nothing in the orchestrator subsequently reacts to the human's decision — there is no inbound callback surface. Approval-to-publish is currently a **manual, out-of-band step**. The code says so plainly: Phase 2b and Phase 3 of the QA feedback design are *"DELIBERATELY NOT implemented … both need a new inbound API surface … that does not exist anywhere in this codebase yet."* (§14.F4)

5.4 Process D — Failure, Retry and Cascade



Two distinct dead-letter reasons exist on purpose, so an operator reading `task_transitions` can tell *"we tried 3 times and gave up"* (`dead_lettered`) from *"we never tried, it was already impossible"* (`dependency_dead_lettered`). The cascade check is **one hop only** — deliberately, because wave-by-wave propagation reaches every descendant on the next check without a recursive lineage walk.

5.5 Process E — Nightly Analytics & Reporting



The KPI set is small and deliberate — four numbers that answer *"is this working and what is it costing?"*:

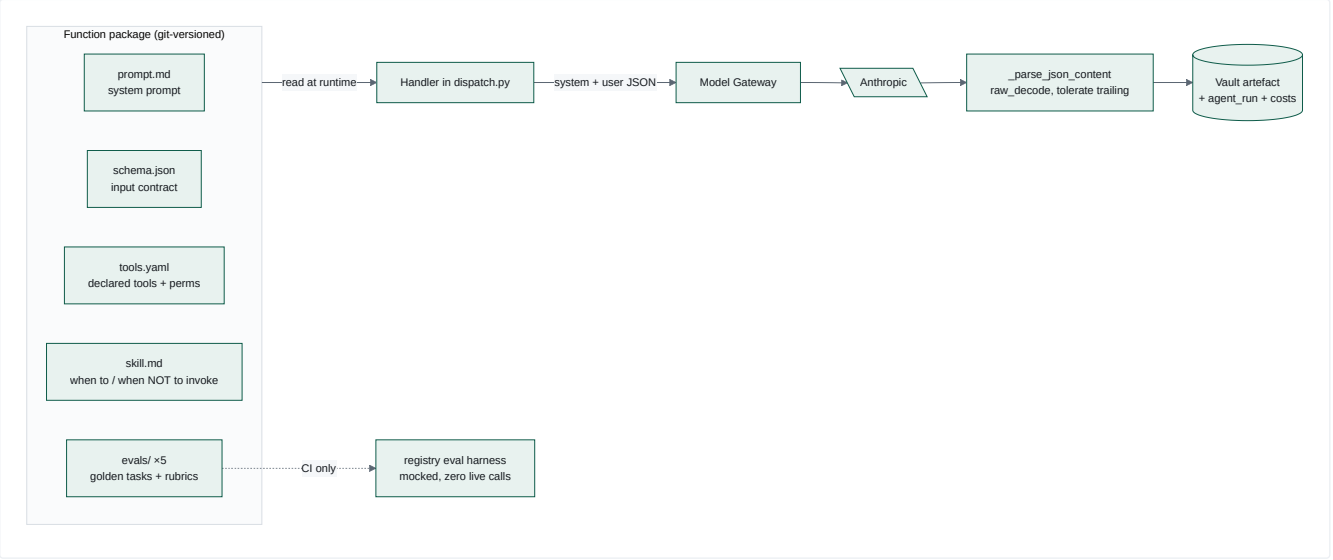
KPI	FORMULA	SOURCE
Engagement rate by post archetype	Σ(reactions+comments+shares+clicks) / Σ(impressions), grouped by (source, archetype)	`buffer_post_metrics` ∪ `linkedin_metrics`
Publishing reliability	scheduled vs. actually-published	`scheduled_posts`
Cost per accepted asset	Σ(costs.usd) / count(approved assets)	`public.costs` → `agent_runs` → `campaigns` ; `assets.approval_state`
Vault utilisation	access counts by object class	`vault_internal.utilisation_daily`

SECTION 6

Agent Architecture

6.1 What counts as an agent here

There is no long-running autonomous agent loop in this system. An "agent" is a **function package** — a versioned bundle of prompt, input schema, tool declarations, skill description and golden evals — invoked by an orchestrator handler as a **single, stateless completion** through the Model Gateway. The agency lives in the *graph*, not in any individual agent.



6.2 The wired agents

Every row below was traced from `DISPATCH_TABLE` through its handler to the `prompt.md` it reads. Revision 2 adds the eleven scanners (§6.3), function 17 (source scout) and the month-end reporter; the fourteen below are the content-producing core.

AGENT	FUNCTION ID	PURPOSE	TRIGGER (TASK_TYPE)	MODEL	KEY INPUTS	OUTPUTS → STORED	DECISIONS IT MAKES
Market Intelligence Director	`09-market-intelligence-director`	Scan a topic, return ≥3 attributed market signals	`ingest-signals`	`claude-haiku`	`fetch_sources.yaml` topic/horizon + 4 fetched page bodies via mcp-web	`signals` row (payload JSONB)	Which items qualify as signals; pillar tagging; confidence level; drops unattributable items
Brief Composer	`brief.compose`	Render signals into a full + executive brief	`draft-brief`	**none** — deterministic	Ancestor `vault_signal_id`	2 `briefs` rows	none (pure rendering)
Brand Steward QA	`02-brand-steward-qa`	Six-check compliance verdict	`qa-review` , `qa-review-brand-steward`	`claude-sonnet`	draft text, `client_references` , channel	verdict → `result_ref` ; `agent_run`	pass/fail; which violation codes apply
Fact-Check Verdict	`48-fact-check-verdict`	Verify every proof point traces to a closed source list	`qa-review-fact-check`	`claude-sonnet`	draft text	verdict → `result_ref`	traceable vs. fabricated
LinkedIn Post Writer	`42-linkedin-post-writer`	One CFO-audience LinkedIn post from one proof point	`draft-content`	`claude-sonnet`	pillar, proof point, campaign UTM	`assets` (linkedin_post)	copy, CTA, hook
Content Planner	`plan.compose`	Rotate the week's pillar	`plan-content-monday`	**none** — deterministic	`CONTENT_PILLARS` list	`result_ref` (pillar)	pillar rotation
Research Brief Writer	`41-research-brief-writer`	Turn signals into a cited research brief	`draft-research-brief`	`claude-sonnet`	week's signals	`briefs` row	which claims have citable sources
Client Advocacy Harvester	`26-client-advocacy-harvester`	Harvest advocacy quotes against consent + permission register	`draft-client-advocacy-harvest`	`claude-sonnet`	consent fixture, permission register	`result_ref`	whether a quote is usable / must stay client-free
Insight-to-Story Editor	`39-insight-to-story-editor`	Narrative story draft	`draft-insight-to-story`	`claude-sonnet` (2048 tok)	research brief, pillar	`assets` (linkedin_post)	narrative framing
Executive Ghostwriter	`43-executive-ghostwriter`	Executive-voice piece, never fabricating an opinion	`draft-executive-ghostwrite`	`claude-sonnet` (2560 tok)	research brief	`assets` (linkedin_post)	voice, what the exec did/didn't say
Carousel Post Writer	`45-carousel-post-writer`	Multi-slide carousel + Canva Bulk Create CSV manifest	`draft-carousel-post`	`claude-sonnet` (2560 tok)	research brief proof points	`assets` (carousel_post)	slide breakdown
Newsletter Writer	`46-newsletter-writer`	Owned-channel newsletter digest	`draft-newsletter`	`claude-sonnet` (3584 tok)	research brief	`assets` (newsletter)	subject line, section selection
Case Study Writer	`47-case-study-writer`	Case study, client-free unless CLEARED	`draft-case-study`	`claude-sonnet` (4096 tok)	research brief + cleared advocacy quote	`assets` (case_study)	whether a client may be named
Content Repurposer	`52-content-repurposer`	Derivative short formats from newsletter/case study	`draft-content-repurpose`	`claude-sonnet`	source draft selection	`assets` (derivatives)	which source, which target formats

Common to every LLM agent:

ASPECT	BEHAVIOUR
Prompt/instructions	`functions/<id>/prompt.md` , read from `FUNCTIONS_DIR` at call time — never inlined in Python
Tools	Declared in `tools.yaml` ; only function 09 actually reaches a tool at runtime (mcp-web `fetch_url`). The rest declare read-only lookups that are implemented as *prompt instructions plus deterministic post-checks*, not runtime tool calls — **an important nuance** (§14.A1)
Data sources	Vault via HTTP; `docs/positioning.md` , `docs/permission-register.yaml` as prompt-referenced sources
Error handling	Exception → `DispatchError` → `_retry_or_dead_letter` → 3 strikes → `dead_lettered` + alert. JSON parse failure logs a 4000-char preview (the only place a raw response is ever persisted)
Human approval	Required only at the publish boundary, via Gatekeeper levels 1/2. Drafting is level 3 (auto-approved, audited); analysis is level 4
Where outputs go	`agent_runs.output` (JSONB) + a typed artefact row + `task_state.result_ref` — three places, deliberately

6.3 The eleven S10 scanners — wired since revision 1

At revision 1 these eleven packages had complete definitions (prompt, schema, tools, skill, 5 evals each) and a task in `daily-signal-loop.yaml` , but **no entry in** `DISPATCH_TABLE` , so every one was a `legacy_task_pass_through` no-op:

`10-competitor-discovery-scanner` · `11-competitor-change-monitor` · `12-competitive-positioning-analyst` · `13-competitor-content-performance-scout` · `16-microsoft-fabric-ecosystem-scout` · `18-01` … `18-06 vertical-intel-*` · `25-competitive-response-strategist`

All eleven are now registered, and the way they are registered is worth knowing:

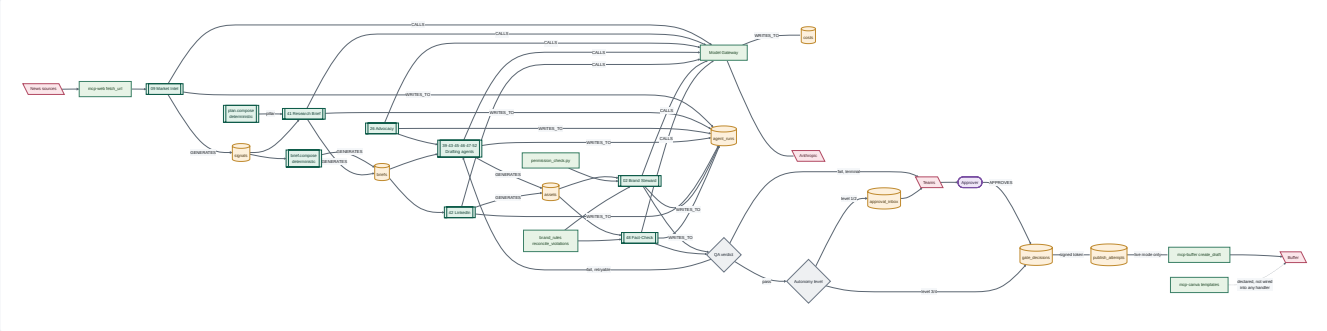
```
SCANNER_TASKS: dict[str, tuple[str, str, str]] = {
    # task_type: (function_id, default profile_id, agent_name)
    "competitor-discovery-scan": ("10-competitor-discovery-scanner", ...),
    ...
}
DISPATCH_TABLE = { **SCANNER_HANDLERS, "ingest-signals": ..., ... }
```

One factory (`_make_scanner_handler`) builds all eleven from a table, and they enter `DISPATCH_TABLE` as a **dict spread**. A grep for `"task-type":` in `DISPATCH_TABLE` therefore misses them and reports eleven false no-ops — resolve `SCANNER_TASKS` before drawing any conclusion. Their scan scope is data, in `functions/_shared/scan-profiles.yaml` .

Deliberately **not** written straight to `opportunity_cards` : the eleven share three listening scopes, so the same event legitimately surfaces several times. De-duplication is `dedupe-signal-cards` ' job, and card rows are written only after it runs (`dispatch.py` 's `SCANNER_TASKS` header comment).

6.4 Agent Interaction Graph

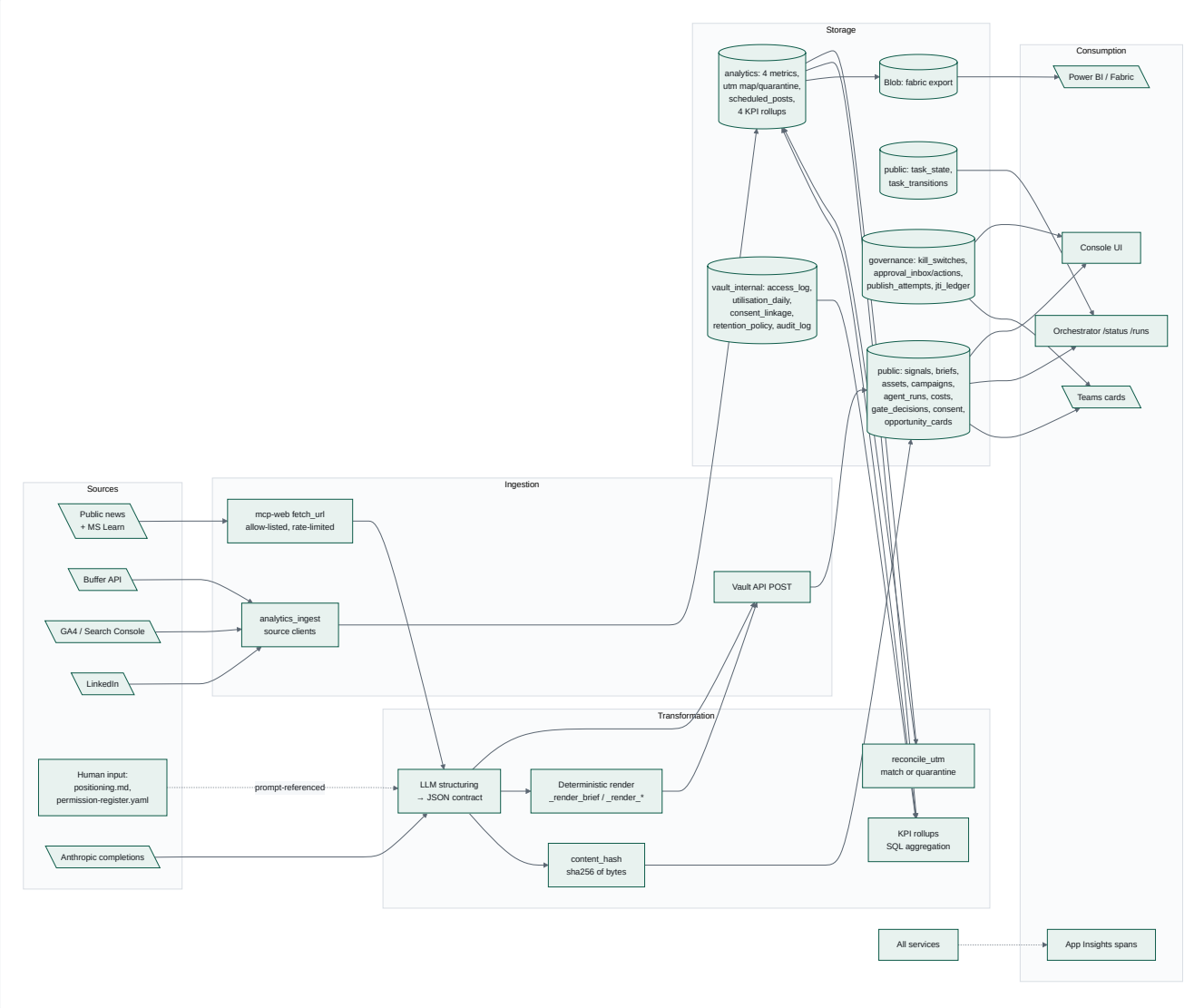
Distinguishing agents ⟦double⟧, tools, databases, external services and humans.



SECTION 7

Data-Flow Architecture

7.1 Source → Ingestion → Transformation → Storage → Processing → Consumption → Output

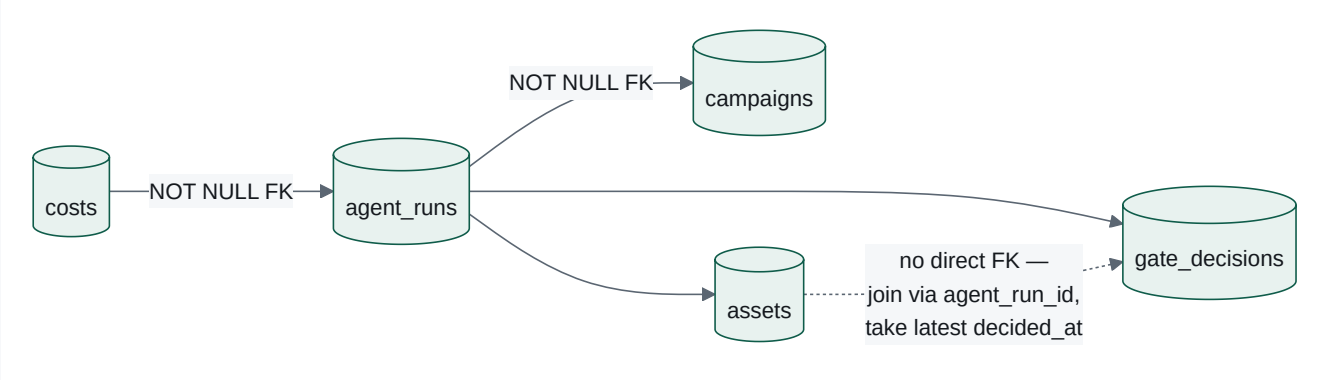


7.2 Major datasets and ownership

One Postgres Flexible Server, four schemas. **Ownership** = the single component permitted to write.

SCHEMA	TABLE	OWNER (WRITER)	READERS	NOTES
public	campaigns	Vault API	Orchestrator, Console, analytics	Top of the cost roll-up chain
public	signals	Vault API (via ingest handler)	draft-brief handler, Console	JSONB payload from function 09
public	opportunity_cards	Vault API	Console	**Declared in the frozen schema; no writer found in any handler** — Unknown / Requires Confirmation
public	briefs	Vault API	drafting handlers, QA, Console	
public	agent_runs	Vault API	Gateway (budget lookup), Publisher, Console, analytics	NOT NULL FK → campaigns
public	assets	Vault API	QA, Publisher, Console	version + approval_state + self-referencing predecessor
public	gate_decisions	Gatekeeper **and** Model Gateway	Publisher, Console	Append-only; two independent writers (§14.D2)
public	costs	Model Gateway metering.py	Gateway budget, analytics, Console	3 rows per completion: usd, tokens, ms
public	consent_register	Vault API	Vault consent gate	POPIA s11/s69-shaped
public	task_state	Orchestrator only	Orchestrator, Console	result_ref JSONB is the inter-task data bus
public	task_transitions	Orchestrator only	Console, operators	CHECK-constrained closed reason vocabulary
governance	kill_switches	Console toggle route	Gatekeeper, Publisher (uncached, every call)	
governance	approval_inbox	Gatekeeper	Approval app, Console	single-use link token, 24h TTL
governance	approval_actions	Approval app	Console, audit	all 4 click outcomes recorded
governance	publish_attempts	Publisher only	Console, audit	one row per branch, including refusals
governance	jti_ledger	Publisher only	Publisher	replay prevention; durable, never in-memory
analytics	4 metrics tables	analytics-ingest	rollups, Power BI	
analytics	utm_campaign_map / utm_quarantine	analytics-ingest	reconciliation	
analytics	4 kpi_rollup_*	analytics-ingest	Fabric export, Power BI	upsert-idempotent
vault_internal	access_log → utilisation_daily	Vault	utilisation KPI	pure telemetry, no personal data
vault_internal	consent_linkage	Vault	audit	links an accepted write to the consent that permitted it
vault_internal	retention_policy	Vault	retention sweep	
vault_internal	audit_log	Vault	audit	rejection + deletion paths only

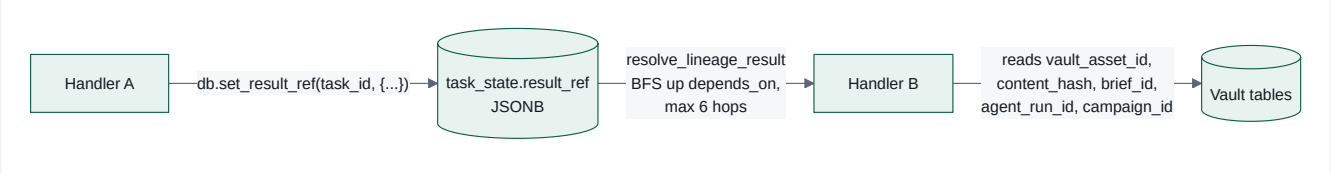
7.3 The unbroken cost chain



costs → agent_runs → campaigns is a deliberate unbroken FK chain enabling per-campaign cost roll-up. assets and gate_decisions deliberately have **no direct FK** to each other; the documented join convention is assets.agent_run_id = gate_decisions.agent_run_id ORDER BY decided_at DESC LIMIT 1, which is authoritative precisely because gate_decisions is append-only.

7.4 The inter-task data bus

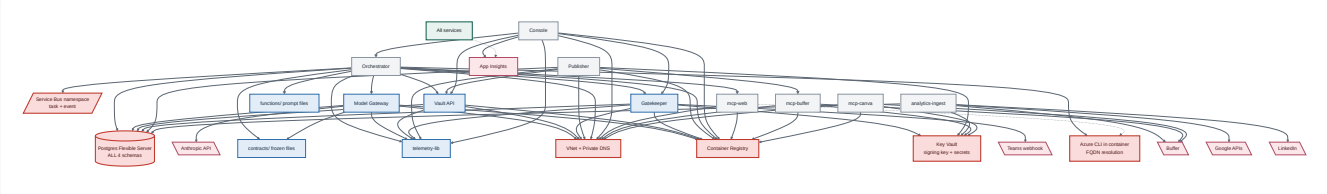
The mechanism by which one task hands work to the next is worth stating explicitly, because it is not obvious from any single file:



`resolve_lineage_result` performs a **breadth-first walk up the `depends_on` ancestry** until it finds a task carrying a non-null `result_ref`. This is what lets it transparently walk *past* a pass-through no-op — `draft-brief`'s immediate predecessor is `score-signals` (a no-op with no `result_ref`), so the walk continues to `ingest-signals` two hops back. One mechanism, no per-loop special-casing.

SECTION 8

Dependency Graph



8.1 Critical dependencies and single points of failure

RANK	COMPONENT	BLAST RADIUS IF IT FAILS	MITIGATION PRESENT?
1	**Postgres Flexible Server**	Total. All 4 schemas — domain data, task state, governance audit, analytics — sit on one server. Orchestrator can't transition tasks, Gatekeeper can't decide, Publisher can't check replay, Console shows nothing.	None architecturally. `/health` endpoints deliberately avoid DB round-trips so probes stay green — which *masks* the outage rather than mitigating it.
2	**Service Bus namespace**	No work starts and no task progresses. Heartbeats are lost (Logic Apps do not retry into a durable store).	Local double exists for tests only.
3	**Model Gateway**	Every LLM agent halts. It is the *only* permitted path to a provider.	Single-purpose service, small surface; per-replica cache means no shared-state failure.
4	**Key Vault**	No new gate tokens can be signed → nothing can be authorised to publish.	Publisher can read the public key from an env-threaded secret, so *verification* survives; *issuance* does not.
5	**Vault API**	Every handler fails: agents can't create runs, assets or briefs.	None.
6	**Azure CLI availability inside containers**	`resolve_live_fqdn` shells out to `az containerapp show`. If the CLI is missing, unauthenticated, or slow, service discovery returns `None` and clients fail — *by design it never fabricates a hostname*.	Env-var override (`CMOS_*_BASE_URL`) wins; results are memoised per process.
7	**Anthropic API**	All content generation stops; deterministic handlers (`brief.compose`, `plan.compose`) still work.	Provider registry makes a second provider a data + one-registration change.
8	**The orchestrator worker task**	It is a single `asyncio.Task` inside the FastAPI process. If its startup raised, `worker_task = None` and the API still serves `/health` 200 — **the system looks healthy and does nothing**.	`worker_loop_start_failed` is logged at WARNING only.

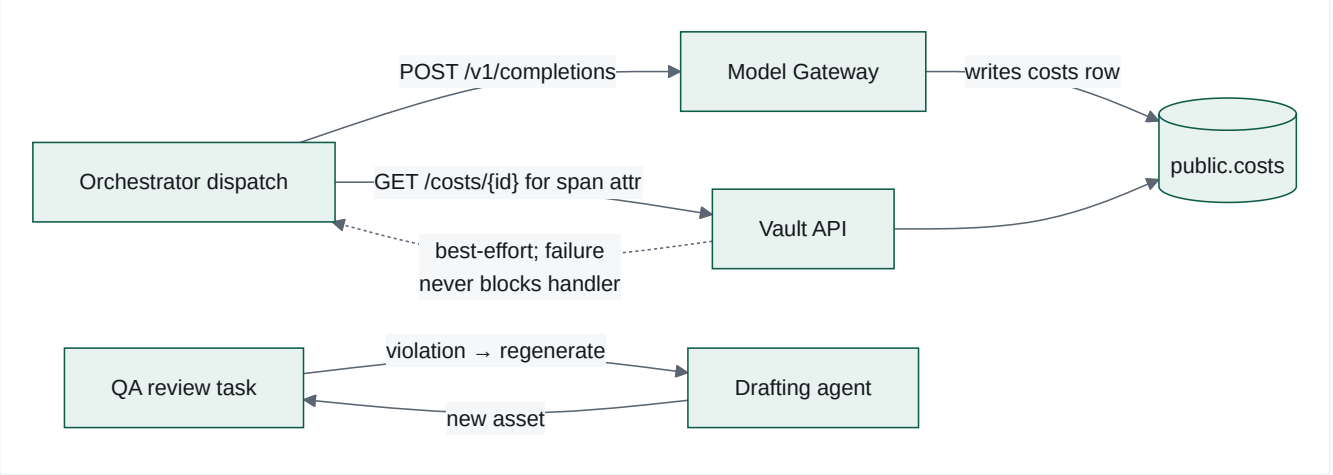
8.2 Shared services and highly connected components

COMPONENT	IN-DEGREE	WHY IT MATTERS
Postgres	9 services	Everything
telemetry-lib	7 services	Installed editable from source in every image; a breaking change touches every service simultaneously
contracts/	4 services read at **runtime** (not just build)	Gateway reads `openapi.yaml` and `redaction-rules.yaml`; Orchestrator reads `loop-definition.schema.json`. Requires per-image staging + `CONTRACTS_DIR`
`functions/` prompt files	Orchestrator (runtime) + registry (CI)	Requires per-image staging + `FUNCTIONS_DIR`
Vault API	4 callers	
`AGENT_NAME_LOOP_PROOF` constant	Duplicated in orchestrator + publisher	Kept honest by a cross-service test — a deliberate coupling made explicit
`kill_switch.py`	Duplicated in gatekeeper + publisher	Kept honest by `test_kill_switch_parity.py`

8.3 Circular dependencies

No import-level cycles exist between services — each is independently packaged, and the two duplicated modules (`kill_switch.py`, verifier constants) exist precisely *to avoid* a shared library that would create one.

Two **runtime** cycles do exist and are intentional:



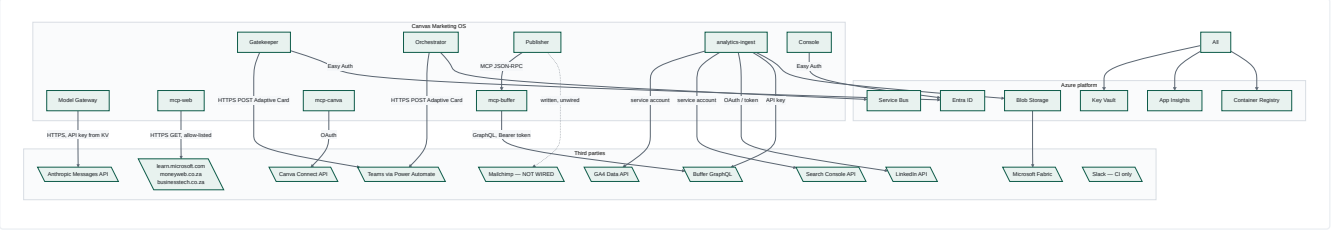
The QA ↔ drafting cycle is bounded at `MAX_QA_RETRY_ATTEMPTS = 10` and serialised by a Postgres advisory lock. The cost-lookup cycle is explicitly best-effort — a failure only leaves the telemetry span's `cost` attribute at `0.0`.

8.4 Lazy imports as deliberate decoupling

`dispatch.py` imports `teams_notify` **inside** functions, `state_machine` imports `dead_letter.emit_alert` inside `record_failure`, and `worker.handle_task_message` imports `dispatch` inside the function body. These break what would otherwise be import cycles within the orchestrator package — a real pattern worth knowing before refactoring.

SECTION 9

Integration Map



9.1 Integration register

EXTERNAL SYSTEM	PURPOSE	CONNECTION	AUTH	DATA SENT	DATA RECEIVED	FREQUENCY / TRIGGER	FAILURE HANDLING
Anthropic Messages API	All content generation and QA verdicts	HTTPS via `providers/anthropic.py`, behind Model Gateway	API key from Key Vault	System prompt (static file) + user JSON (redaction-scanned)	Completion text + token usage + stop_reason	Every agent task, ~40+ calls/day	`httpx.HTTPStatusError` → 500 `PROVIDER_ERROR`; empty content logged WARNING; task retried 3× then dead-lettered
Buffer (GraphQL)	Social scheduling	mcp-buffer + a separate Publisher client	Bearer token, `BUFFER_API_KEY`	`channel_id`, `text` **only**	Post id, queue list	Live-mode publish; nightly metrics pull	Fixture mode when credential absent; free-tier queue cap (10) checked live before create; `BufferClientError` → refusal row
Canva Connect	Design creation from brand templates	mcp-canva	OAuth (`scripts/oauth_consent.py`)	`template_id` + autofill data	Design id, export URL	**Not invoked by any orchestrator handler today**	Fixture mode default
Microsoft Teams	Approval cards, brief cards, needs-edit / retries-exhausted cards	HTTPS POST to a Power Automate HTTP-trigger URL	URL is the credential (from Key Vault)	Adaptive Card 1.4 JSON with `Action.OpenUrl` deep links	none	On every level-1/2 escalation and QA block	Absent `TEAMS_WEBHOOK_URL` → silent no-op; the inbox row remains the delivery mechanism
GA4 Data API	Landing-page metrics	`ga4_client.py` + `google_auth.py`	Google service account	Property id, date range	Sessions, landing pages, UTM	Nightly	Fixture fallback; failure isolated per source
Search Console API	Organic search metrics	`search_console_client.py`	Google service account	Site URL, date range	Queries, impressions, clicks	Nightly	As above; **no `utm_campaign` column** — excluded from reconciliation by construction
LinkedIn API	Post performance	`linkedin_client.py`	Token	Org id, date	Impressions, reactions, comments, shares	Nightly	Fixture fallback
Microsoft Fabric / Power BI	Executive reporting	Blob container shortcut + `analytics/powerbi/analytics-dataset.json`	Managed identity to blob; Fabric-side shortcut	Validated nightly export JSON	none	Nightly after rollups	Payload self-validated before upload; upload failure reported as a simulated marker in dry-run
Mailchimp	Newsletter send	`publisher/app/esp_client.py`	**Not configured**	—	—	—	**NOT WIRED.** Three non-code prerequisites are documented as blocking: a Mailchimp account, an audience with a physical mailing address, and CRM sync. Approving a newsletter card today sends no email.
Public news sources	Market signal evidence	mcp-web `fetch_url`	none	URL only	Page body, truncated to 2000 chars	Daily, 4 URLs	Per-source failure skipped; if all fail → `DispatchError`; redaction block → drop one source and retry
Entra ID	Human identity	Container Apps Easy Auth	OIDC	—	Principal headers	Every console/approval request	`Return401` on unauthenticated
Slack	Deploy notifications	GitHub Actions reusable workflow	Webhook secret	Deploy status + run URL	none	Per deploy	CI-only; no runtime dependency

9.2 Credential handling

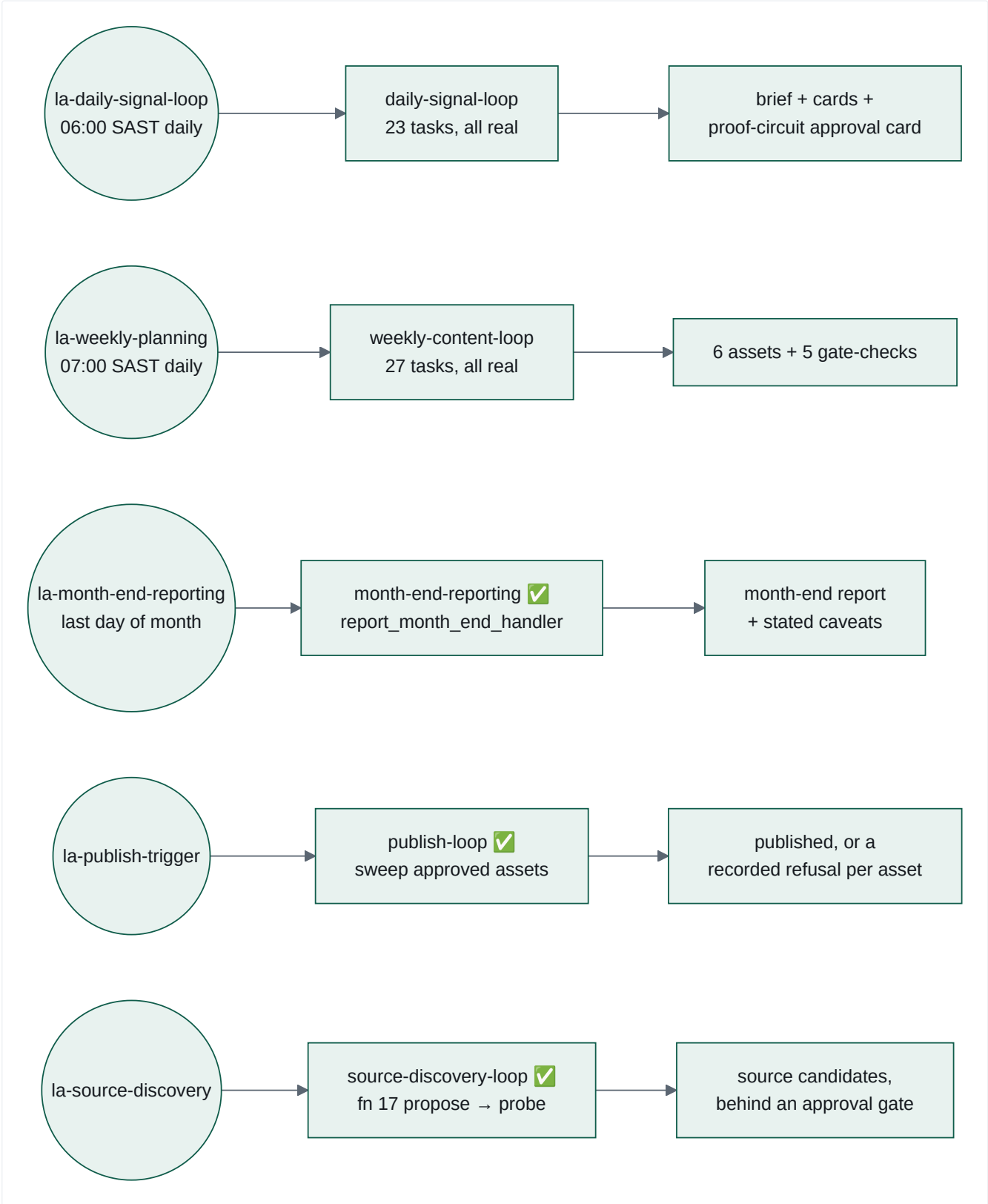
No credential is committed. `docs/credentials-runbook.md` defines the Key Vault naming convention and the POPIA cross-border transfer considerations per foreign-hosted provider. Container Apps secrets are base64-encoded on the way in to defend against the platform's `$$ → $` collapse. Infra deploys use **OIDC federated identity — no client secrets** (`.github/workflows/deploy-infra.yml`). The one committed keypair is `services/registry/keys/dev-signing-key.*`, and every use of it prints a WARNING to stderr on every run.

SECTION 10

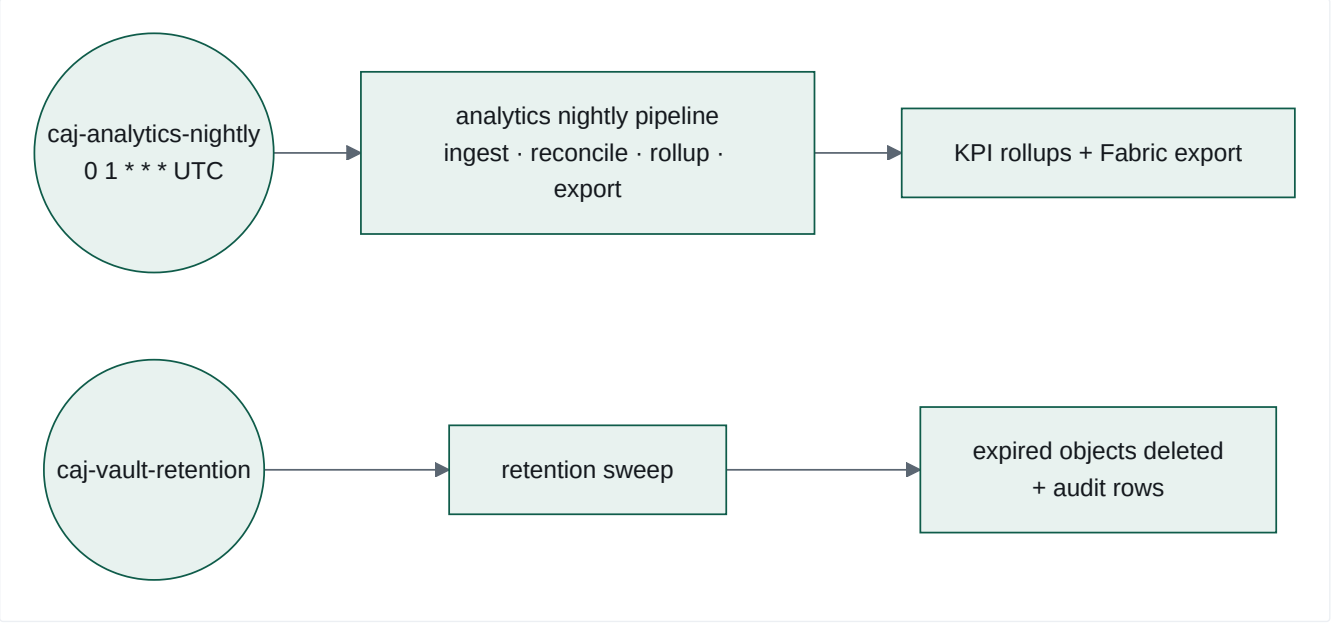
System Trigger Map

Everything that can initiate activity, traced **Trigger → Process → Component/Agent → Output**.

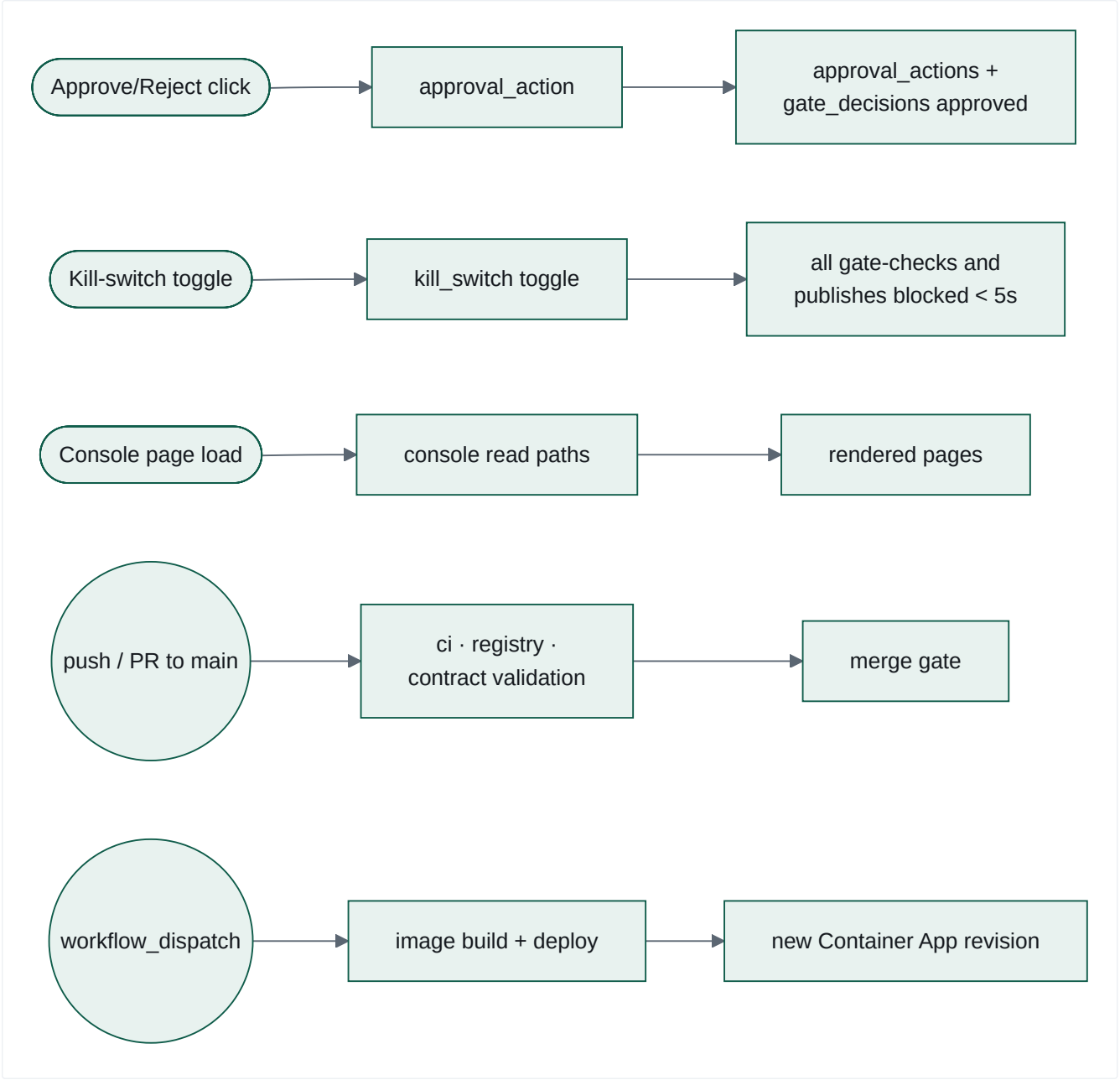
① **Loop triggers** — Logic Apps that publish a heartbeat onto the `event` queue for the orchestrator to decompose:



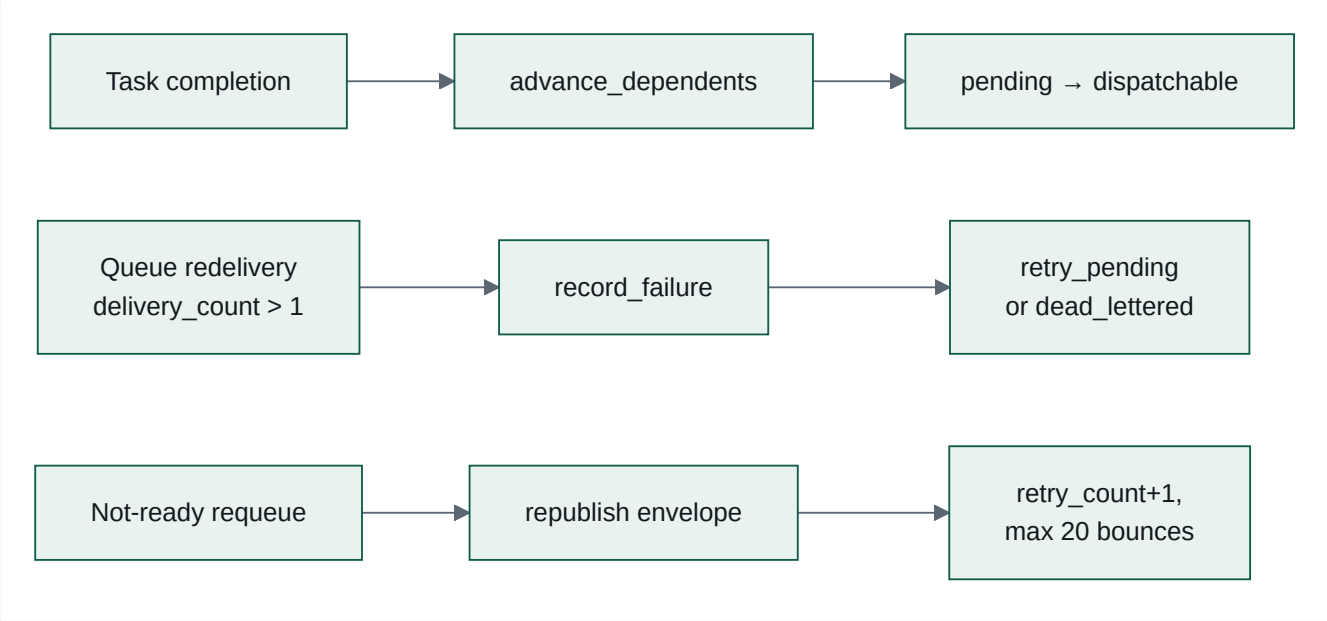
② **Container Apps Jobs** — scheduled compute that bypasses the orchestrator entirely:



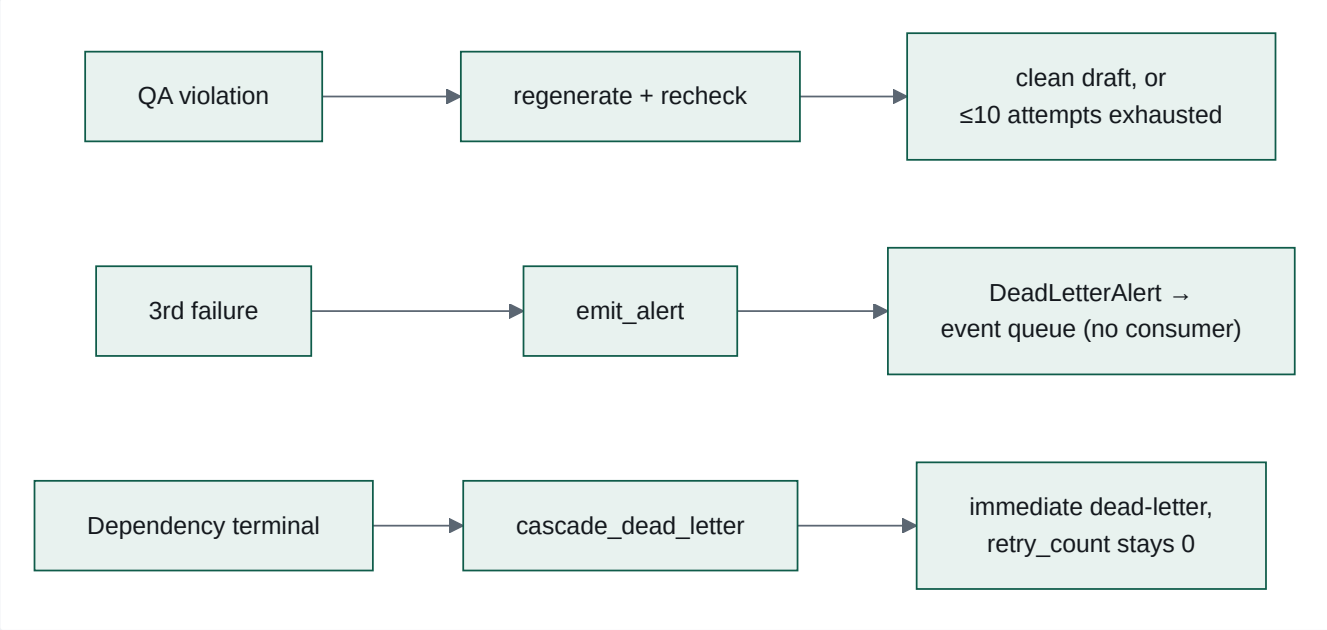
③ Human and developer-initiated triggers:



④ System-internal triggers — task lifecycle:



⑤ System-internal triggers — failure and correction:



10.1 Trigger register

TRIGGER	TYPE	CADENCE	MECHANISM	INITIATES	NOTES
`la-daily-signal-loop-trigger`	Scheduled	Daily 06:00 SAST	Logic App → HTTP POST to Service Bus `event` with MSI auth	daily-signal-loop	Sender role only, never Receiver
`la-weekly-planning-trigger`	Scheduled	**Daily 07:00 SAST**	Logic App → `event`	weekly-content-loop	Daily is the deliberate standing cadence (confirmed 17 Aug 2026), not a pending revert. One complete Mon-Fri content cycle per fire — the day-named task ids are dependency-chain names, not a schedule
`la-month-end-reporting-trigger`	Scheduled	Monthly	Logic App → `event`	`month-end-reporting` loop	✅ Fixed since revision 1 — the loop file now exists and `report_month_end_handler` renders a month-end report that states its own caveats.
`la-publish-trigger`	Scheduled	Per Bicep	Logic App → `event`	`publish-loop`	✅ New. Sweeps approved-but-unpublished assets; dry-run by default.
`la-source-discovery-trigger`	Scheduled	Per Bicep	Logic App → `event`	`source-discovery-loop`	✅ New. Function 17 proposes and probes new sources behind its own approval gate.
`caj-analytics-nightly-ingest`	Scheduled job	`0 1 * * *` UTC	Container Apps Job	Full analytics pipeline	Bypasses the orchestrator entirely
`caj-vault-retention-expiry`	Scheduled job	Per Bicep	Container Apps Job → `python -m vault.retention`	Retention sweep	
`caj-orchestrator-migrate`, `caj-governance-migration`, `caj-vault-*-migration`, `caj-analytics-migration`	Deploy job	On deploy	Container Apps Jobs, base64-encoded SQL secret	Schema migrations	
`caj-loop-e2e-smoke`, `caj-governance-smoke`, `caj-mcp-smoke`, `caj-*-smoke`	Deploy job	On deploy	Container Apps Jobs	Post-deploy verification	`caj-loop-e2e-smoke` budget: 40 attempts × 15 s = 600 s
Approve/Reject link click	Human	Ad hoc	GET on external approval app	Approval finalisation	Single-use, 24h TTL
Kill-switch toggle	Human	Ad hoc	Console POST	Global or per-function block	Propagates to the very next decision — no cache
`advance_dependents`	System event	Per completion	Direct SQL	Downstream task readiness	
Service Bus redelivery	System event	On lock lapse	`delivery_count > 1`	`record_failure`	Signature of a crash between receive and complete
Not-ready requeue	System event	Per poll pass	Republish	Deferred dispatch	Bounded at 20
QA violation	Agent decision	Per failed review	Advisory-lock-guarded loop	Regeneration	≤10 attempts
Dead-letter	System event	3rd failure	`emit_alert` → `event` queue	**Nothing consumes it**	Logged as informational only
GitHub push / PR	CI	Per commit	Actions	Lint, contract validation, migration test, registry gates, bundle verification	

SECTION 11

Detailed Component Catalogue

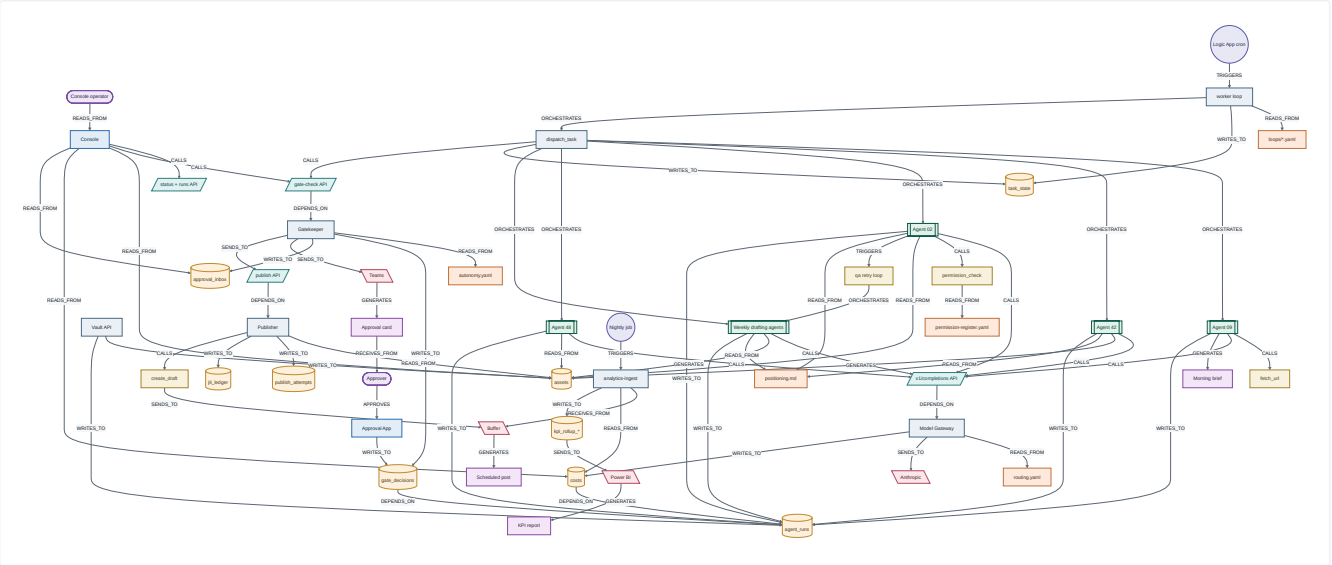
COMPONENT	TYPE	PURPOSE	INPUTS	OUTPUTS	DEPENDENCIES	TRIGGER	DATA USED	CONN COMP
orchestrator/main.py	Service (FastAPI)	HTTP surface + worker lifespan	HTTP	/health, /status, /runs/{ref}, /tasks/{id}/review	db, worker, loop_loader	HTTP request; process start	task_state, task_transitions	Consol suite
orchestrator/worker.py	Background loop	Poll both queues, route messages	Queue messages	Task envelopes, state transitions	Service Bus, db, dispatch	Continuous poll (1 s)	task_state	dispatc state_m produc
orchestrator/dispatch.py	Router + 18 handlers	Execute a task_type	TaskEnvelope, ancestor result_ref	Vault artefacts, gate-checks, result_ref	Gateway, Vault, Gatekeeper, mcp-web, functions/	Called by worker	all Vault tables	every a every p service
orchestrator/state_machine.py	Module	Retry / backoff / dead-letter / cascade	task_id, db	Transitions + alerts	db, dead_letter	Handler failure	task_state	worker dead_l
orchestrator/decompose.py	Module	Loop + heartbeat → task list	LoopDefinition, HeartbeatEvent	Task dicts with uuid5 ids	—	Heartbeat	—	worker
orchestrator/loop_loader.py	Module	Load + validate loop YAML, enforce acyclicity	YAML files	LoopDefinition	contracts/orchestrator/loop-definition.schema.json	Process start	—	main, v
orchestrator/db.py	Data access	All task-state SQL incl. advisory locks	SQL params	Rows, lock connections	Postgres	Every handler	task_state, task_transitions	everyth orchest
orchestrator/teams_notify.py	Integration	Brief / needs-edit / retries-exhausted cards	Card fields	HTTP POST	TEAMS_WEBHOOK_URL, CMOS_CONSOLE_BASE_URL	Handler events	—	Teams,
orchestrator/brand_rules.py	Module	Drop QA false positives	violations, draft text	reconciled violations	—	Every QA verdict	—	QA har
model-gateway/completion.py	Pipeline	The 5-stage completion pipeline	Request dict	(status, body)	routing, redaction, budget, caching, metering, providers	HTTP POST	costs, gate_decisions, agent_runs	all ager
model-gateway/redaction.py	Firewall	Block PII-shaped content pre-provider	payload, exemptions	RedactionResult	redaction-rules.yaml	Every completion	—	comple gate_d
model-gateway/budget.py	Policy	Per-agent_name daily budget	agent_run_id, tier	(tier, state)	budgets.yaml, costs	Every completion	costs, agent_runs	comple routing
model-gateway/caching.py	Idempotency	One compute per task_ref	task_ref, closure	Response + hit flag	—	Every completion	—	comple
model-gateway/metering.py	Accounting	3 cost rows per completion	tier, tokens, latency	cost_id	—	Successful completion	costs	Vault, a
model-gateway/providers/registry.py	Extension point	provider name → adapter	name	Provider instance	—	Per completion	—	anthrop
gatekeeper/routers/gate_check.py	Decision engine	Autonomy ruling + token issuance	GateCheckRequest	GateCheckResponse + 1 audit row	policy_loader, kill_switch, approval_inbox, signer	HTTP POST	gate_decisions, approval_inbox, kill_switches	Orches Publish
gatekeeper/app/tokens.py	Crypto	Mint RS256 gate tokens	decision id, hash, function_id	JWT	signer (Key Vault or local)	Approved decision	—	Publish
gatekeeper/app/approval_inbox.py	Workflow	Single-use, TTL-bounded approval links	decision + preview	inbox row, urls, card	Teams client	Level 1/2 escalation	approval_inbox	Approv Consol
gatekeeper/app/kill_switch.py	Control	Uncached block check	conn, function_id	BlockStatus	Postgres	Every decision	kill_switches	gate_ch
publisher/routers/publish.py	Enforcement	Verify, refuse or publish	PublishRequest	1 publish_attempts row	verifier, jti_ledger, kill_switch, vault_lookup, buffer_client	HTTP POST	governance.*, Vault assets	mcp-bu
publisher/app/verifier.py	Crypto	Alg-pinned JWT verification	token, public key	claims or VerificationError	PyJWT only — **imports nothing from app**	Every publish	—	publish parity t

COMPONENT	TYPE	PURPOSE	INPUTS	OUTPUTS	DEPENDENCIES	TRIGGER	DATA USED	CONN COMP
publisher/app/jti_ledger.py	Replay guard	Durable single-use jti	jti	consumed / already-seen	Postgres	Every publish	jti_ledger	publish
vault/routers/objects.py	CRUD	9 object types + consent gate	HTTP	Domain rows	consent, storage, audit	HTTP	public.* , vault_internal.*	every a Consol
vault/retention.py	Job + route	Retention-class expiry sweep	date	Deletions + audit	storage	Job or HTTP	retention_policy	retentic
vault/rollup.py	Accounting	Access log → daily utilisation	date	Upserted rows	—	Per GET + rollup	access_log , utilisation_daily	analytic
analytics-ingest/cli.py	Job entrypoint	run / nightly subcommands	day, source flags	Row counts, JSON	all ingest modules	Container Apps Job	analytics.*	Fabric,
analytics-ingest/utm.py	Transform	Reconcile or quarantine UTM	new rows	matched / quarantined	utm_campaign_map	Nightly	analytics.*	rollups
analytics-ingest/fabric_export.py	Export	Assemble + self-validate payload	day	Validated dict	fabric-nightly-export.schema.json	Nightly	4 KPI tables	blob_w
mcp/common/mcp_common/protocol.py	Framework	MCP JSON-RPC over HTTP	JSON-RPC	initialize / tools/list / tools/call	FastAPI	HTTP POST /mcp	—	all 3 M
mcp-web/app/tools.py	Tool	Allow-listed, rate-limited fetch	url	body	allow-list, rate limiter	tools/call	—	functio
mcp-buffer/app/dispatch.py	Tool	Read queue + create draft	channel_id, text	Buffer response	Buffer GraphQL	tools/call	—	Publish
mcp-canva/app/dispatch.py	Tool	Template-locked design creation	template_id + data	design/export	Canva API	tools/call	—	**no orc caller**
console/app/routes_reads.py	UI	7 read pages	HTTP + Easy Auth	HTML	services layer	HTTP	all schemas via APIs	Orches Vault, Gateke Insight
console/app/routes_write.py	UI	Kill-switch toggle only	HTTP + same-origin	State change	Gatekeeper client	HTTP POST	kill_switches	Gateke
registry/build_registry.py	CI tool	Reproducible signed manifest	functions/	registry.json + .sig	Ed25519 signing	CI	—	CI gate
registry/eval_harness.py	CI tool	Golden evals, mocked	eval fixtures	Pass/fail per rubric	stub gateway	CI	—	CI gate
registry/safety_suite.py	CI tool	Deterministic brand checks	markdown	Violation lines	regex + lexicon	CI + declared tool	—	functio tools.y
telemetry-lib	Library	Typed, enum-constrained spans	attributes	OTel spans	OpenTelemetry	Import	—	7 servi
scripts/validate_contracts.py	CI tool	Contract correctness + freeze guard	contracts/	Pass/fail	.frozen-v1.sha256	CI	—	CI
functions/task-worker/function_app.py	**Stub**	Azure Function health route only	HTTP	{"status":"ok"}	—	HTTP	—	**nothin** real co the orc worker

SECTION 12

System Relationship Graph

Typed nodes, typed edges. This exposes relationships that reading any single file would not reveal.



12.1 Non-obvious relationships this graph exposes

RELATIONSHIP	WHY IT ISN'T OBVIOUS	CONSEQUENCE
Model Gateway `WRITES_TO` `gate_decisions`	`gate_decisions` reads like a Gatekeeper-owned table	Two independent services append to the same audit table with different `decided_by` prefixes (`gatekeeper:policy` , and the gateway's redaction/budget deciders). Auditing "all decisions" must query both.
Budget is keyed on `agent_name` **, resolved from** `agent_run_id`	Budgets look campaign-scoped	A budget breach for one agent affects every loop that uses it. `agent_name` is set by the handler, and proof-circuit runs rewrite it to `loop-proof-circuit` — so the proof circuit has its **own** budget bucket.
`result_ref` **is the real inter-task data bus**	It's a JSONB column on a state table	Task coupling is via JSONB key names, not typed contracts — a rename breaks a downstream handler silently.
`resolve_lineage_result` **walks *past* no-op tasks**	Looks like a simple parent lookup	The 17 daily-loop no-ops are invisible to downstream handlers, which is why the loop still "works" despite them.
Publisher reads `agent_runs.agent_name` **to force dry-run**	Publisher looks purely token-driven	A data value in the Vault overrides a runtime environment flag — a safety property enforced from the data side.
`request-approval` **completes before any human decides**	Named like a blocking gate	The task graph proceeds; there is no wait-for-approval state.
Console reads App Insights directly	Console looks like a Postgres reader	Trace pages depend on the App Insights query API, a fourth data source beyond the three schemas.
`brand_rules.reconcile_violations` **can overturn an LLM verdict**	QA looks model-authoritative	A deterministic post-check drops model false positives before the pass/fail decision — the model is advisory on those codes.

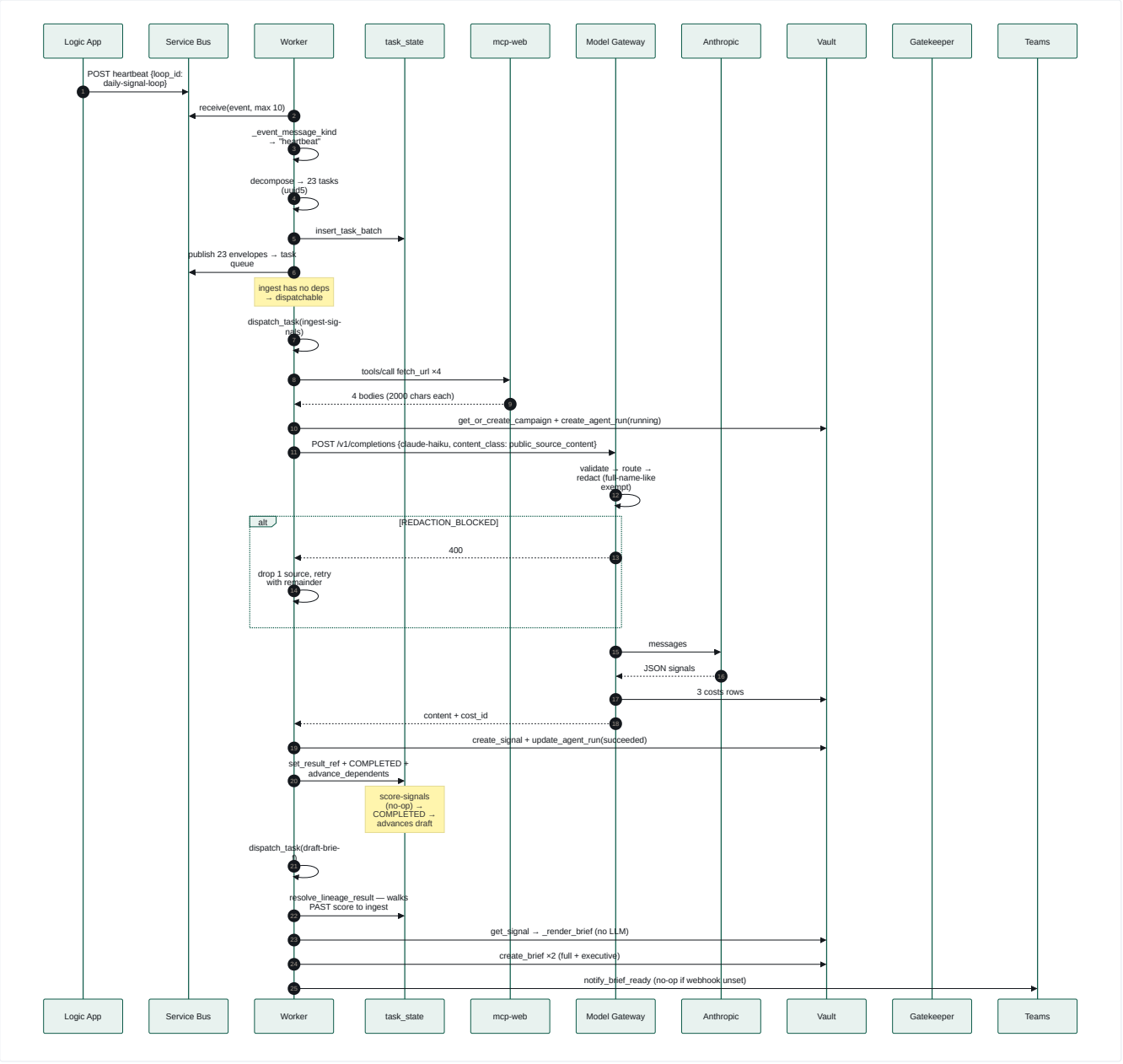
SECTION 13

Critical End-to-End Journeys

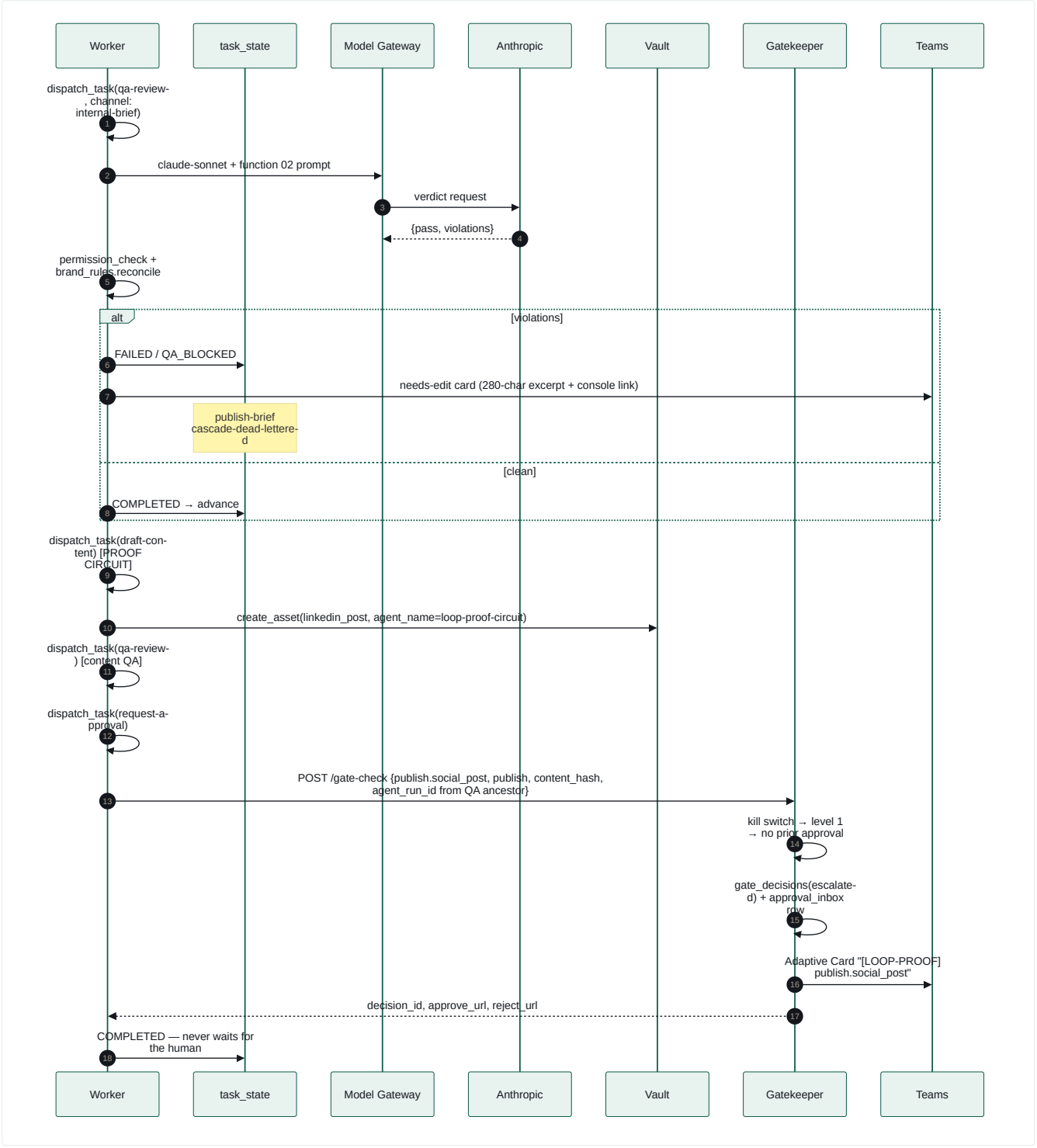
13.1 Journey 1 — A market signal becomes an approval card

The full daily path, every component and decision.

Part A — trigger, signal ingestion and brief composition.



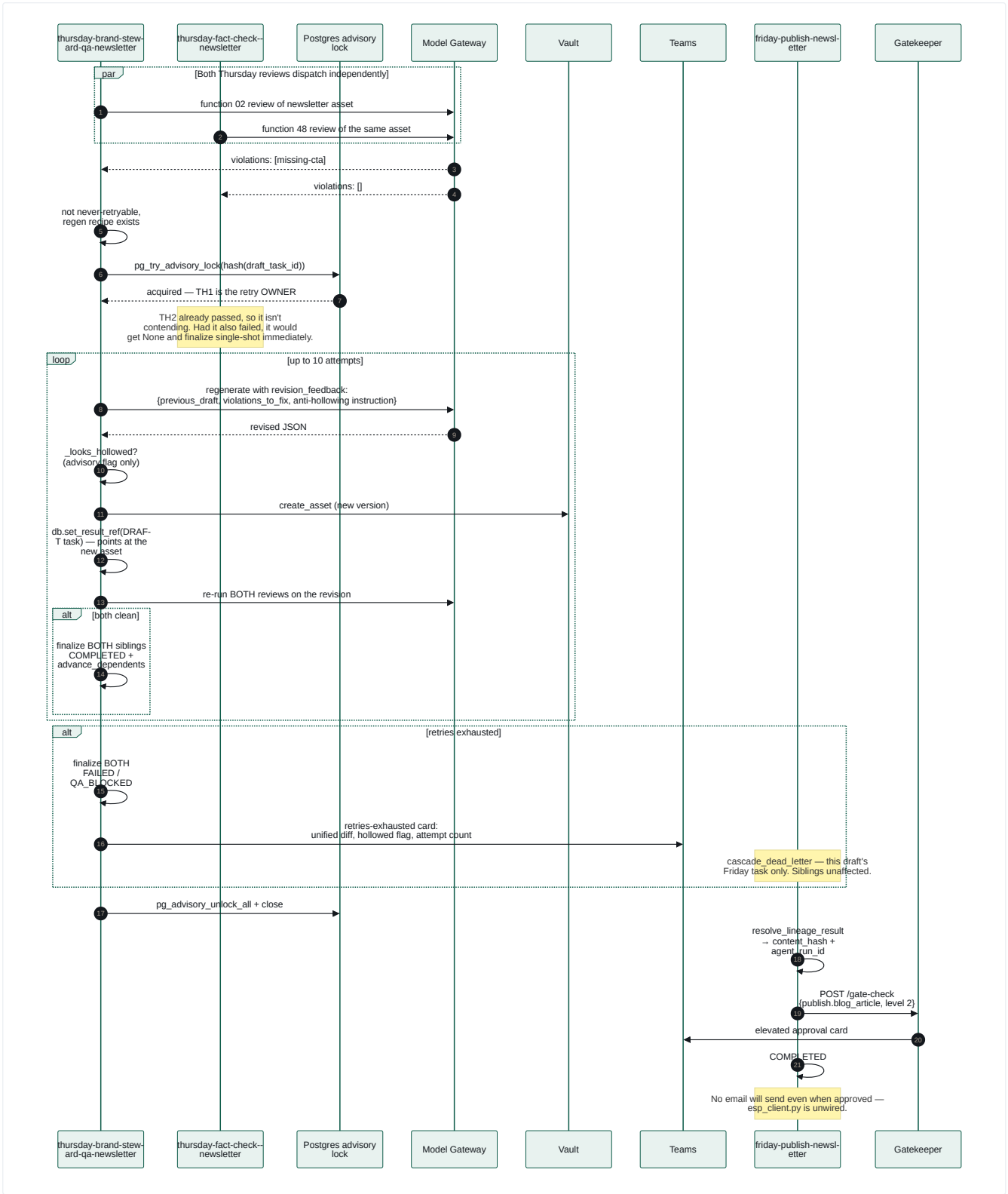
Part B — QA gate, proof circuit and approval card. Continues from the brief written above.



Final outcome: a morning brief in the Vault, a Teams brief card (when the webhook is configured), and a clearly-tagged `[LOOP-PROOF]` approval card whose eventual approval is hard-wired to dry-run.

13.2 Journey 2 — A weekly draft fails QA, self-corrects, and reaches Buffer

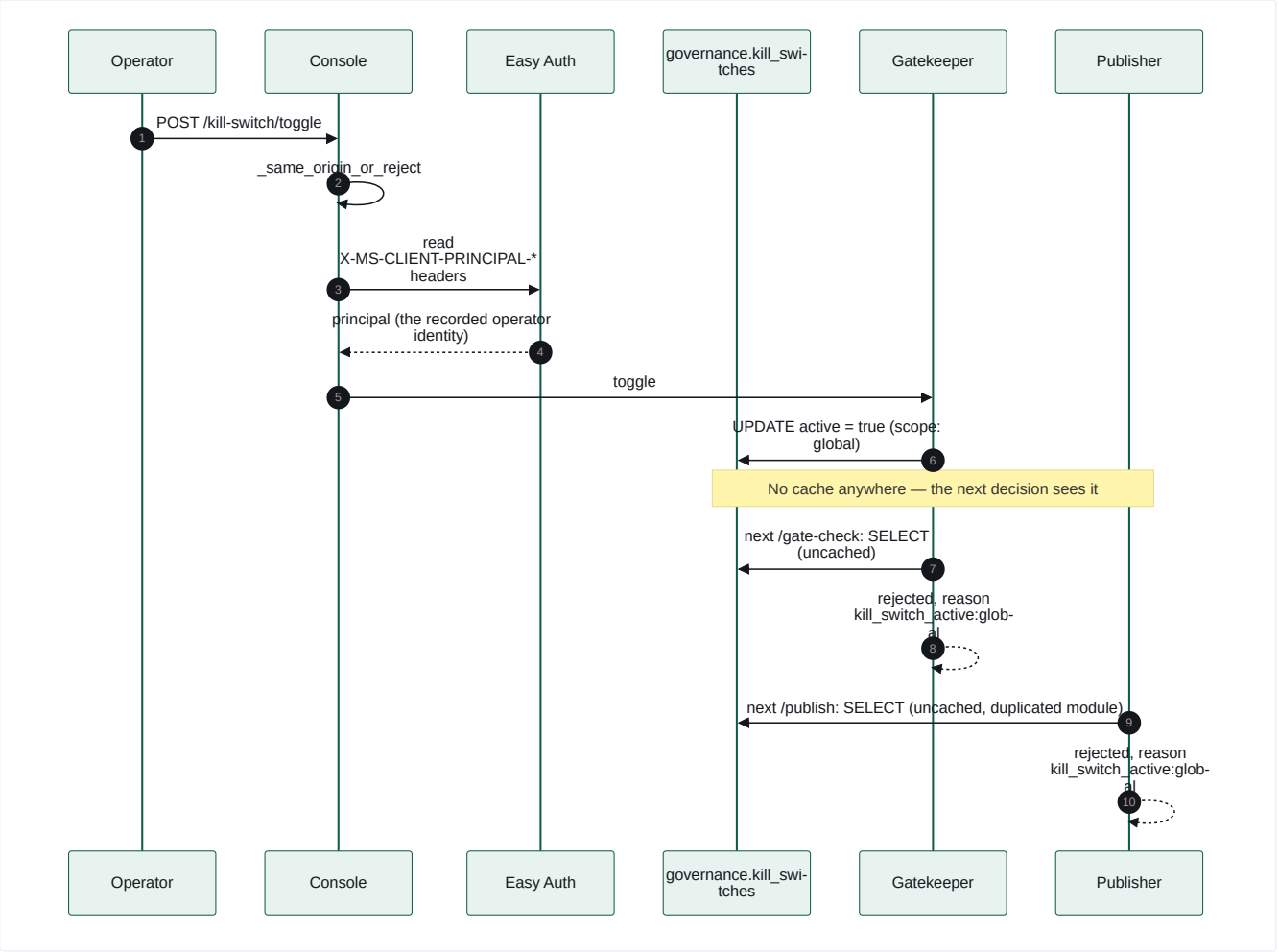
The most intricate control flow in the system.



Three subtleties worth calling out:

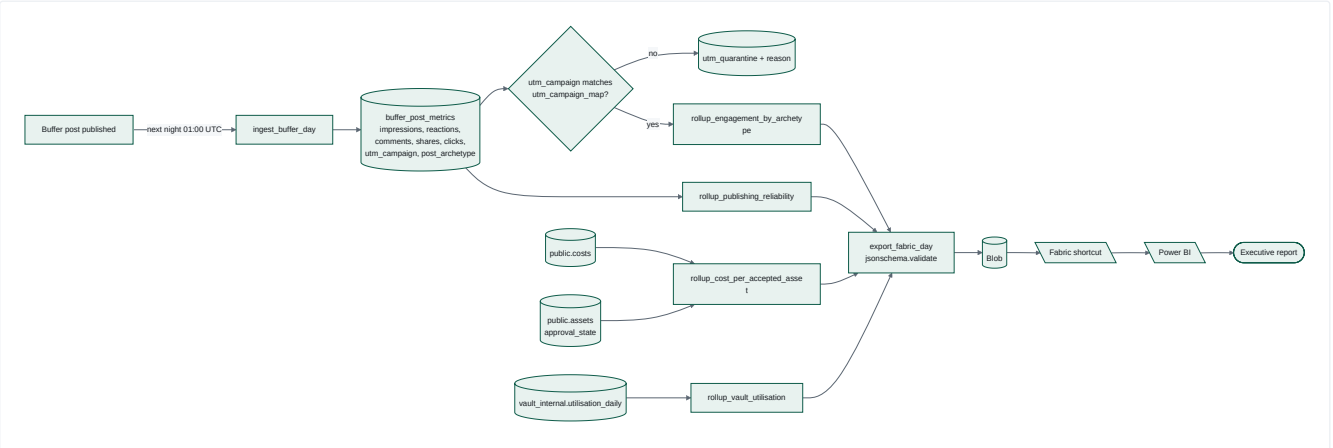
1. The retry loop **rewrites the draft task's** `result_ref`, so Friday's lineage walk resolves the *corrected* asset, not the original.
2. The owner finalises **both** sibling tasks, including one it doesn't own — the only place in the codebase where a handler writes another task's terminal state.
3. A losing sibling emits a "needs edit" card that the winner may supersede seconds later. Documented and accepted: *"a possible duplicate/stale-looking Teams notification is far cheaper than a stuck task handler."*

13.3 Journey 3 — An operator kills everything



Propagation is bounded by the next database read, not by a cache TTL. The module is duplicated across two services that share no library, and `test_kill_switch_parity.py` loads **both files** and asserts identical behaviour across the full scope/function_id matrix.

13.4 Journey 4 — A published post becomes a KPI



This is the only feedback path from published output back to a decision-relevant number — and it is currently **advisory**: nothing in the orchestrator reads `kpi_rollup_*` to change future content decisions (§14.F5).

SECTION 14

Architecture Findings

Facts observed in the codebase are marked 🔍. Interpretations and recommendations are marked 💬.

14.0 What changed between revision 1 and revision 2

`main` advanced 59 commits between `5c8ee07` and `e17a157`. Re-verifying every revision-1 finding against the current tree:

#	REVISION-1 FINDING	STATUS NOW	EVIDENCE
F1	17 of 23 daily-loop tasks were no-ops	✅ **Closed.** 0 of 23. All eleven scanners wired via a factory, plus `score-signals`, `dedupe-signal-cards`, `competitive-response-strategize`, both brief rollups and `publish-brief`	`dispatch.py` `SCANNER_TASKS` / `DISPATCH_TABLE`
F4	Monthly trigger fired at a loop file that didn't exist	✅ **Closed.** `month-end-reporting-loop.yaml` + `report_month_end_handler`	`loops/month-end-reporting-loop.yaml`
F7	Approval → publish was a manual, out-of-band step	✅ **Closed.** `publish-loop.yaml` sweeps approved-but-unpublished assets, re-checks each approval, mints a token on a second `/gate-check`, and posts the exact approved bytes to Publisher	`loops/publish-loop.yaml`, `publish_approved_assets_handler`
F10	`opportunity_cards` had no writer	✅ **Closed.** Written by scoring/dedupe, read back by planning and the brief	`vault_client_ext.create_opportunity_card`, `list_opportunity_cards`
P4	Monday planning was `week % 5`, blind to data	✅ **Closed.** Reads recent scored signals, picks the top pillar, falls back to the rotation only when there is no evidence — and records which of the two it used	`plan_content_monday_handler`, `_recent_scored_signals`, `_top_pillar`
—	*(new)* Scoring thresholds were implicit	✅ Now reviewed data	`functions/_shared/scoring-policy.yaml`
—	*(new)* Source list was a fixed allowlist	✅ `source-discovery-loop` + function 17 propose/probe sources behind their own approval gate	`loops/source-discovery-loop.yaml`
—	*(new)* Measurement had nothing to join on	✅ `record_scheduled_post()` writes `analytics.scheduled_posts`; campaigns register in `analytics.utm_campaign_map`	`orchestrator/db.py:551,582`
F2	Nightly analytics loop is documentary, not executed	⬜ **Unchanged, by design.** Still 7 no-op task types; the real pipeline is `caj-analytics-nightly-ingest`	loop file header
F3	Weekly trigger fires daily, not Monday	✅ **Closed — as intended behaviour, not a revert.** Confirmed 17 Aug 2026 that daily *is* the standing cadence. The `TEMPORARY` marker and the dead commented-out weekly block are removed and the header now states the ruling. **This document's revision-1 framing was wrong**: it read a stale comment as evidence of a defect and costed it as ~7× overspend. The cadence was a deliberate product decision throughout	`weekly-planning-trigger.bicep` header
F8	Newsletter ESP written but unwired	❌ **Still open.** No `esp_client` import in any publish router	`publisher/app/routers/`
F12	Fact-check prompt self-declares "not approved policy"	❌ **Still open.** Header unchanged, and it still gates publication	`functions/48-fact-check-verdict/prompt.md:3`
F5	Performance never feeds decisions	⚠️ **Partly.** Archetype engagement is now read — but only by `_render_month_end_report`. Planning is driven by *signal* scores, never by *published-post performance*	`dispatch.py:3962`, `db.py:413`
F6	Dead-letter alert has no consumer	❌ **Still open, and worse than recorded.** The branch is still `informational only today`, and there are **no Azure Monitor alert rules anywhere in `infra/`** — so a dead-lettered task emits an alert onto a queue nobody reads, into a log nobody is paged on	`worker.py:426-437`; no `metricAlerts` / `scheduledQueryRules` in IaC
F9	mcp-canva has no orchestrator caller	❌ **Still open, and materially more serious than "unused".** It is a **fully deployed Container App** — own managed identity, Key Vault and ACR role assignments, and two live OAuth secrets (`CANVA_CLIENT_ID`, `CANVA_CLIENT_SECRET`) — built and shipped by `deploy-mcp.yml`, and **invoked by nothing**. Function 45 still emits a `canva_bulk_create_csv` manifest into every carousel asset that no handler consumes	`infra/main.bicep:957,994,1031,1114-1130`; `functions/45-*/prompt.md:23`; zero callers outside `mcp/`
F11	`functions/task-worker/` is a health-check stub	❌ **Still open, but harmless.** 23 lines, one route. Confirmed **not deployed and not built** — no reference in `infra/` or any workflow. Pure dead scaffolding	the file; absent from `infra/`, `.github/workflows/`

The one attribution gap that survived. `create_draft` still sends exactly two arguments — `channel_id` and `text` — so `analytics.post_archetype` still has **no writer anywhere**. It is read by the month-end report and by `db.py`'s engagement query, but nothing populates it: the headline KPI still groups on a column the system never fills. The other two join keys (`scheduled_posts`, `utm_campaign_map`) were closed in revision 2; this is the remaining third, and it is now the single highest-value small fix in the codebase.

Complexity moved sharply the other way. `dispatch.py` went from **2,890 to 6,920 lines** — a 2.4× increase in the file that was already the hardest to reason about. Everything in §14.3's C1 entry applies with more force, and §15's R5 (split `dispatch.py`) is now the highest-priority maintainability item rather than a nice-to-have.

14.1 Architectural strengths

#	STRENGTH	EVIDENCE
S+1	🔍 **Governance is genuinely architectural, not procedural.** Autonomy levels, asymmetric signed tokens with algorithm pinning, a durable replay ledger, canonical-JSON hash binding, single-use TTL-bounded approval links, and an append-only decision trail. 💬 This is materially stronger than most systems of this size.	`contracts/gate-token/spec.md` , `publisher/app/verifier.py` , `gatekeeper/app/approval_inbox.py`
S+2	🔍 **Policy-as-data throughout.** Routing, budgets, autonomy, loops, fetch sources and redaction rules are all reviewable YAML. A model upgrade is one reviewed line.	`policy/*.yaml` , `loops/*.yaml`
S+3	🔍 **Fail-closed defaults are consistent.** Unlisted autonomy pair → blocked. Unlisted client → blocked. Failed Vault lookup → refuse. Missing register file → empty index, not "allow all".	`autonomy.yaml` , `permission_check.py`
S+4	🔍 **Every branch leaves exactly one audit row**, including refusals — with distinct machine-readable reason strings.	`routers/publish.py` , `routers/gate_check.py` docstrings
S+5	🔍 **The comment culture is exceptional.** Nearly every non-obvious decision carries a dated incident writeup naming the symptom, the root cause and the reasoning. 💬 This is the single biggest asset for a new engineer, and it is what made this reverse-engineering tractable.	throughout, e.g. `redaction.py` INCIDENT 1–3
S+6	🔍 **Isolation via graph shape, not handler filtering.** The round-34 restructure fixed a real cascade bug by changing the dependency graph rather than adding conditional logic.	`weekly-content-loop.yaml` header
S+7	🔍 **Contract freezing with a hash guard**, plus additive-extension discipline (packing `function_id` / `content_hash` into an existing claim rather than breaking the freeze).	`contracts/.frozen-v1.sha256` , `gate-token/spec.md` addendum
S+8	🔍 **Reproducible, signed function registry** — no timestamps, no absolute paths, CRLF-normalised hashes, deterministic Ed25519, verified byte-identical twice in CI.	`registry/build_registry.py` , `.github/workflows/registry.yml`
S+9	🔍 **Telemetry is redaction-aware by construction** — a closed attribute-key enum and a 200-char structural rejection, mirroring the queue's no-free-text rule.	`telemetry_lib/attributes.py`

14.2 Facts: gaps between declared and actual behaviour

> **Read with §14.0.** This table is preserved as it stood at revision 1, because the reasoning behind each finding is still the clearest statement of *why* it mattered. F1, F4, F7 and F10 are now **closed** — see §14.0 for what closed them. F3, F8 and F12 remain open exactly as written.

#	FINDING	EVIDENCE	💬 IMPACT
F1	🔍 **17 of 23 `daily-signal-loop` tasks have no handler.** All 11 competitive/vertical-intelligence scanners, plus dedupe, response-strategise, both brief rollups, `score-signals` and `publish-brief`, fall through to `legacy_task_pass_through` — `RUNNING → COMPLETED`, producing nothing. Their function packages (prompts, schemas, evals) exist and are complete.	`DISPATCH_TABLE` vs. `daily-signal-loop.yaml`, verified programmatically	The loop reports success daily while ~74% of its declared work is a no-op. An operator reading `/status` sees 23 completed tasks. This is the highest-value finding in the document.
F2	🔍 **`nightly-analytics-ingest-loop.yaml` is documentation, not execution.** All 7 task types are unhandled; the real work runs in a separate Container Apps Job. The file's own header says so.	loop file header, `nightly-ingest-job.bicep`	Honest, but it means a loop file in `loops/` may or may not be executable — a reader can't tell without checking `DISPATCH_TABLE`.
F3	🔍 **The weekly trigger fires daily.** `frequency: Day, interval: 1`.	`weekly-planning-trigger.bicep`	✅ **Resolved as intended, 17 Aug 2026 — this finding was a misreading.** A `TEMPORARY` comment awaiting a revert that never came is indistinguishable in source from a defect, and this document called it one. It is the standing cadence: one complete content cycle every morning for review. The marker is gone and the file now says so. **Lesson worth keeping: a stale comment is evidence of a stale comment, not of intent.** The real residual risk is a live one — see F13.
F4	🔍 **`la-month-end-reporting-trigger` targets a loop that does not exist.** No `month-end-reporting` file in `loops/`. The Bicep comment acknowledges the heartbeat will be "logged and skipped".	`month-end-reporting-trigger.bicep`, `ls loops/`	A monthly no-op that logs a warning. Deployed infrastructure with no effect.
F5	🔍 **No feedback loop from published performance to decisions.** At revision 2 archetype engagement is read by the month-end report, and planning is now evidence-led — but off *signal* scores, not off *what actually performed*.	`dispatch.py:3962`; `plan_content_monday_handler`	The measurement subsystem still informs humans, not the machine.
F6	🔍 **`DeadLetterAlert` has no consumer**, and **no alert rule exists in IaC**. The worker logs it as "informational only today"; nothing pages anyone.	`worker.py:426-437`; no `metricAlerts` / `scheduledQueryRules` under `infra/`	A task can exhaust its retries and die with the only trace being a log line nobody is watching. This is the observability gap (§14.7 O2) with a concrete failure attached.
F7	🔍 **Approval → publish is not automated.** `request-approval` completes on gate-check response; there is no inbound callback surface. Phases 2b and 3 of the QA design are explicitly deferred for exactly this reason.	`dispatch.py` F-QA-RETRY-LOOP block	The last mile is manual.
F8	🔍 **`esp_client.py` is written but unwired**; `publish_newsletter_handler` requests approval and stops. The code says so explicitly.	`esp_client.py`, `dispatch.py` module docstring	Approving a newsletter card sends no email.
F9	🔍 **mcp-canva is deployed but invoked by nothing.** A live Container App with its own identity, Key Vault + ACR role assignments and two Canva OAuth secrets, shipped by `deploy-mcp.yml` — with zero callers outside `mcp/` itself. Function 45 still emits a `canva_bulk_create_csv` manifest nothing consumes.	`infra/main.bicep:957,994,1031,1114-1130`; `functions/45-*/prompt.md:23`	Not merely "carousel production is manual". This is standing compute cost plus a live third-party credential held by a service with no consumer — surface that exists only to be attacked. Either wire it or decommission it.
F10	🔍 **`opportunity_cards` has no writer.** The table is in the frozen schema and routed by the Vault API, but no handler creates a row.	`contracts/vault-schema/schema.sql`, `dispatch.py`	"Opportunity scoring", named in the README, is not implemented.
F11	🔍 **`functions/task-worker/function_app.py` is a health-check stub** whose docstring says the real consumer "is implemented in a later wave" — it was, in the orchestrator. Confirmed **not deployed and not built**: no reference in `infra/` or any workflow.	the file; absent from `infra/`, `.github/workflows/`	Harmless at runtime, but it implies an Azure Functions tier that does not exist, and every reader has to work that out. Delete it.
F12	🔍 **`48-fact-check-verdict/prompt.md` is a self-declared unapproved first draft** — yet it gates real content reaching Buffer and the newsletter.	the prompt's own header	An unreviewed policy is in the production critical path.
F13	🔍 **The daily cadence will breach Buffer's queue cap the moment publishing goes live.** Each cycle requests 4 social posts (`friday-schedule-social-buffer-*` × 4). Daily × 4 = ~28 queued posts/week against a free-tier cap of **10**, enforced by a live `list_queue` count in Publisher.	`weekly-content-loop.yaml`; `publisher/app/config.py:BUFFER_FREE_TIER_QUEUE_CAP`	💬 Currently masked: `PUBLISHER_DRY_RUN` defaults true and is set nowhere in infra, so nothing is queued today. It fails safe when it does bite — a `buffer_queue_cap_exceeded` refusal row, not a crash — but it will refuse silently from roughly day 3 of live publishing. Decide before flipping the flag: a paid Buffer tier, fewer posts per cycle, or accepting that the cap throttles output.

14.3 Complexity hotspots

#	HOTSPOT	🔍 FACT	💬 ASSESSMENT
C1	`dispatch.py` at **6,920 lines** (2,890 at revision 1) holds routing, 28+ handlers plus an 11-handler factory, the QA retry loop, scoring, scanning, dedupe, month-end reporting, lineage resolution, JSON parsing and 4 exception classes.	measured	The single hardest file to reason about. Handler bodies are formulaic and near-duplicated; the retry loop alone is ~450 lines.
C2	**Dispatch readiness is a five-way branch** — dispatchable / already-terminal / dependency-terminal / not-ready / unregistered — each with a different recovery.	`dispatch_task`	Correct and well-documented, but understanding it requires reading four exception docstrings totalling ~150 lines.
C3	**The queue-bounce mechanism.** All tasks are published up front; not-ready tasks bounce up to 20 times. The `NOT_READY_MAX_REQUEUES = 20` comment shows it was tuned against an observed ~14 s inter-requeue interval to fit inside a 600 s smoke budget.	`worker.py`	💬 The bound is empirically fitted to current graph width and replica count. A wider loop or a slower stage could exceed it and dead-letter healthy tasks. This is the most fragile numeric constant in the system.
C4	**Three overlapping identifiers** for one unit of work: `task_id` (uuid5), `envelope.agent_run_id` (a *synthetic* uuid5 that is never a real row), and the real `agent_run_id` inside `result_ref`. Confusing these caused a live FK-violation bug.	`request_approval_handler` ROUND 34 docstring	💬 A naming-level trap that has already cost one production incident.
C5	`content_class` **exemption sprawl.** Four separately-authorised call sites now set `public_source_content`, each requiring its own recorded sign-off.	`redaction.py` INCIDENT 2/3	💬 The governance around it is exemplary; the trend is the concern.

14.4 Duplicate functionality

#	🔍 DUPLICATION	WHY IT EXISTS	💬 VIEW
D1	The six brand-safety rules exist **twice**: as prose in `02-brand-steward-qa/prompt.md` (model-judged) and as regex/lexicon in `registry/safety_suite.py` (deterministic, CI-only).	Different execution contexts	Real drift risk — the deterministic suite is never run against live drafts.
D2	`gate_decisions` has **two writer services** with different `decided_by` conventions.	Gateway needed an audit sink for redaction/budget rulings	Reasonable reuse; makes "list all decisions" queries subtle.
D3	`kill_switch.py` duplicated in gatekeeper + publisher.	The two services share no library	Deliberate, parity-tested. Acceptable.
D4	`AGENT_NAME_LOOP_PROOF` duplicated across services.	Same reason	Deliberate, test-enforced.
D5	Two independent Buffer clients (`mcp-buffer/app/dispatch.py` , `publisher/app/buffer_client.py`) and two `resolve_live_fqdn` implementations.	Same reason	The FQDN one is a genuine copy-paste — `orchestrator/clients/azure_fqdn.py` already generalises it.
D6	`qa_review_handler` and `_single_draft_qa_review` are structural siblings with near-identical bodies.	Deliberately kept separate; documented	💬 ~200 lines of parallel logic that must be changed in lockstep.

14.5 Bottlenecks, fragility and single points of failure

#	ISSUE	🔍 EVIDENCE	💬 RISK
B1	**One Postgres server holds all four schemas.**	`infra/main.bicep`	Total-outage SPOF; no read/write separation; analytics rollups compete with the transactional hot path.
B2	**The `task_ref` idempotency cache is process-local**, explicitly "out of scope" for multi-replica consistency — while the orchestrator runs `maxReplicas: 3` .	`caching.py` docstring	Concurrent replicas can double-spend on the same `task_ref` .
B3	**Console Vault search fetches full lists and filters client-side**, flagged in-code as interim.	`console/app/services.py`	Degrades linearly with Vault growth.
B4	`resolve_live_fqdn` **shells out to** `az` at runtime from inside containers.	`clients/azure_fqdn.py`	A CLI/auth/latency problem becomes a service-discovery outage. Memoised per process, which also means a redeployed dependency isn't re-resolved until restart.
B5	**The worker is a single asyncio task inside the API process.** If startup fails it is set to `None` , `/health` still returns 200.	`main.py` lifespan	Silent total stall that looks healthy.
B6	`_retry_or_dead_letter` **sleeps 2 s inside the handler path** and re-invokes the same handler in-process.	`worker.py`	Ties up a worker slot; the retry is not queue-mediated.
B7	**The QA retry loop can issue up to 30 model calls for one draft** (10 regenerations × 1 regen + 2 reviews).	`_run_qa_retry_loop`	A pathological draft can consume a large share of the daily budget. Budget breach only *downgrades* the tier — it doesn't stop the loop.
B8	**Handler retries are not idempotent**, acknowledged in the code: *"a handler that partially wrote to Vault before failing is not guaranteed idempotent on retry (e.g. a duplicate signal/agent_run row is possible)."*	`worker.py:_retry_or_dead_letter`	Duplicate Vault rows on retry.

14.6 Security observations

#	OBSERVATION	🔍 EVIDENCE	💬 ASSESSMENT
SEC1	🔍 **Service Bus Standard SKU, no private endpoint, public endpoint reachable.**	`docs/accepted-risks.md`	Formally accepted with three compensating controls (`disableLocalAuth` , TLS 1.2+, metadata-only envelopes) and a documented production hardening path. Well handled.
SEC2	🔍 **No authentication between internal services.** Orchestrator, Vault, Gatekeeper, Publisher and Gateway accept any in-VNet caller. `/gate-check` authenticates nothing beyond a well-formed UUID.	`orchestrator/main.py` route docstring, `gate_check.py`	Consistent and documented, and the `smoke.governance_cycle` entry was *removed* precisely because it made this exploitable. 💬 The entire security model rests on the VNet boundary holding.
SEC3	🔍 **A committed dev signing keypair** exists at `services/registry/keys/` .	the files	Registry-artefact signing only, never gate tokens; every use prints a stderr WARNING. Acceptable, worth a periodic re-check.
SEC4	🔍 **Buffer channel IDs and org ID are hardcoded** in `publisher/app/config.py` and `weekly-content-loop.yaml` .	those files	Not secrets, but environment-coupled config in source.
SEC5	🔍 **Redaction pattern coverage is acknowledged as incomplete** in its own module docstring.	`redaction.py`	Honest. It is defence-in-depth on top of the consent regime, not the primary control.
SEC6	🔍 **Console `/health` is deliberately unauthenticated** to let Container Apps probes bypass Easy Auth, with a residual-risk note.	`console/app/main.py`	Correctly scoped — returns only `{"status":"ok"}` .

14.7 Scalability, maintainability and observability

#	CONCERN	🔍 FACT	💬 VIEW
SC1	Queue-bounce coordination is O(tasks × poll passes); tuned empirically for today's graph width.	`NOT_READY_MAX_REQUEUES` comment	Won't scale to substantially wider loops.
SC2	`advance_dependents` runs a query per candidate per completion.	`db.py`	Fine at 26 tasks; not at thousands.
SC3	Analytics reads and transactional writes share one server.	`main.bicep`	Contention grows with history.
M1	Model prices are a hand-maintained dict in `metering.py` .	that file	Cost figures silently drift from reality on any provider price change.
M2	Runtime dependence on staged `contracts/` and `functions/` directories via `CONTRACTS_DIR` / `FUNCTIONS_DIR` has caused this bug class **twice** (documented as L-0062).	`orchestrator/config.py`	Each new runtime-read directory is a new instance of the same trap.
M3	`result_ref` coupling is by untyped JSONB key name.	throughout `dispatch.py`	A key rename breaks a downstream handler with no compile- or test-time signal.
O1	🔍 **Config absence is indistinguishable from config error.** Missing `TEAMS_WEBHOOK_URL` , `DATABASE_URL` , App Insights, or a failed worker start all log at WARNING/INFO and continue.	`main.py` , `teams_notify.py`	The degrade-gracefully principle (D6) has an observability cost: there is no "expected-but-absent" alarm.
O2	🔍 **No alerting at all in IaC.** Re-confirmed at revision 2: `infra/` contains no `metricAlerts` , no `scheduledQueryRules` , no action groups — so nothing pages on dead-letters, QA-block rates, budget breaches or a loop that silently stops completing.	no alert rules anywhere under `infra/`	Detection is entirely manual. Combined with F6 (the dead-letter alert nothing consumes) and B5 (a failed worker still serving `/health` 200), the system's most likely failure mode — doing nothing while looking healthy — has no automated detector. Alerts could still exist portal-side outside IaC; if so they are undiscoverable from the repo, which is its own problem.
O3	🔍 A raw model response is persisted **only** on JSON parse failure (4000-char preview). `agent_runs` stays at `running` when a handler fails before `update_agent_run` .	`_parse_json_content`	Failed runs leave a stale `running` row; there is no reaper.

14.8 Areas that are difficult to understand

1. **The five-way readiness branch** in `dispatch_task` (§14.C2).
2. **The three-identifier confusion** around `agent_run_id` (§14.C4).
3. **The advisory-lock ownership transfer**, where one task finalises another's terminal state (§13.2).
4. **Which loop files are executable** — you must diff the YAML against `DISPATCH_TABLE` (§14.F1/F2).
5. **The `content_class` exemption chain** — four call sites, three incidents, one narrow pattern.
6. **Lazy in-function imports** that exist to break cycles (§8.4) and disappear from a static import graph.

SECTION 15

Architecture Improvement Opportunities

💬 **This section is recommendation only. No code was changed.**

> **Revision 2 status. R1 (make declared work provably executed), R3 (close approve → publish)** and the planning half of **R12** have all been implemented on `main` — see §14.0. **R2 (weekly cadence)** is untouched and is now the cheapest open win in the document. **R5 (split `dispatch.py`)** has gone from *High* to the top of the list: the file has grown 2.4× since it was written. The revised order is:
>
> 1. ~~**R2 — weekly cadence**~~ **withdrawn: daily is intended** (17 Aug 2026). See R2 for why this document got it wrong
> 2. **P9a** — thread `utm_campaign` + `post_archetype` through `create_draft`; the last unpopulated join key (§14.0)
> 3. **R5** — split `dispatch.py`, now 6,920 lines
> 4. **R6** — observability, still entirely open (F6 + O2 + B5 compound)
> 5. **F13** — decide the Buffer queue-cap posture before publishing goes live
>
> Everything below is preserved as written at revision 1; the closed items are marked in §14.0 rather than deleted, so the reasoning survives.

CRITICAL

R1 — Make declared work and executed work provably identical ✅ *implemented on `main` (§14.0)*

Current	17 of 23 daily-loop tasks and all 7 nightly-loop tasks silently pass through with no handler (F1, F2).
Problem	The system reports success for work it never performed. An operator cannot distinguish "done" from "skipped" from `/status`.
Proposed	Add a startup assertion in `loop_loader.py`: every `task_type` in every shipped loop must either be in `DISPATCH_TABLE` or carry an explicit `params.passthrough: true`. Fail loudly at startup otherwise. Surface a `handler: real | passthrough` field on `/status` and in the Console.
Benefit	Eliminates the largest declared/actual gap; makes future drift impossible.
Complexity	Low — one loader check, one schema field, one response field.
Dependencies	Requires deciding, per unwired task, whether to implement it or mark it explicitly deferred.
Considerations	The 11 intelligence packages already have prompts, schemas and evals — wiring them is mostly handler plumbing that mirrors `_draft_social_post_handler`. Decide deliberately: implement or delete.

R2 — ~~Restore the weekly trigger to its intended cadence~~ ❌ WITHDRAWN — this recommendation was wrong

The premise was false. Daily *is* the intended cadence, confirmed 17 Aug 2026; the `TEMPORARY` comment it rested on was stale, not a pending revert. Acting on this recommendation would have cut content output to a seventh of what the owner wants.

Kept visible rather than deleted, because the failure mode is instructive and cheap to repeat: **a comment describing intent is not evidence of intent.** Where a schedule, a flag or a threshold looks wrong, confirm against the person who set it before costing it as a defect. The trigger file now states the ruling in its header so the next reader cannot make the same mistake.

The residual risk the cadence *does* carry is real and is tracked separately as **F13** (Buffer queue cap on live publishing).

▶ Original recommendation, as written at revision 1

R3 — Close the approval → publish loop ✅ *implemented on `main` via `publish-loop.yaml` (§14.0)*

Current	`request-approval` completes at gate-check; nothing reacts to the human decision (F7, F8).
Problem	The system automates everything except the final action, so the last mile is manual and the pipeline's value is unrealised.
Proposed	Add a small inbound surface — a `post-approval` task type polled from `approval_inbox`, or an approval-app webhook that publishes an event onto the `event` queue. The orchestrator then dispatches a `publish` task that calls Publisher with the token. Wire `esp_client.py` behind the existing dry-run flag.
Benefit	Completes the product thesis.
Complexity	Medium — new task type, new event kind, Publisher call path.
Dependencies	Mailchimp account + audience + physical address (non-code, documented).
Considerations	Keep dry-run as the default and preserve the proof-circuit forced-dry-run rule.

HIGH

R4 — Replace queue-bouncing with completion-driven dispatch

Current	All tasks are published at decompose time; not-ready tasks bounce ≤20 times (C3, SC1).
Problem	The retry bound is empirically fitted to today's graph width and replica count. A wider loop or slower stage dead-letters healthy tasks. It also generates large volumes of pointless queue traffic.
Proposed	Publish only tasks with no dependencies at decompose time. Have `advance_dependents` publish a task's envelope at the moment it flips to `dispatchable`. Keep `TaskNotReadyError` as a safety net for genuine races, with a much smaller bound.
Benefit	Removes the most fragile constant; eliminates bounce traffic; makes graph width irrelevant to correctness.
Complexity	Medium — `advance_dependents` gains a producer dependency; `NOT_READY_MAX_REQUEUES` becomes a true edge case.
Dependencies	Careful handling of the multi-replica publish race (an insert-guard or a `published_at` column).

R5 — Split `dispatch.py`

Current	2,890 lines mixing routing, 18 handlers, the retry loop and shared helpers (C1).
Problem	Highest-change-rate file in the system; every incident touches it.
Proposed	`dispatch/router.py` (table + readiness), `dispatch/handlers/{ingest,brief,content,qa,approval}.py`, `dispatch/qa_retry.py`, `dispatch/lineage.py`, `dispatch/parsing.py`. Keep `DISPATCH_TABLE` as the single registration point.
Benefit	Reviewability; smaller blast radius; parallel work becomes possible.
Complexity	Medium, mechanical — the existing test suite covers the seams well.
Considerations	Preserve the incident-history comments verbatim; they are the file's highest-value content.

R6 — Make silent degradation observable ⬆ *raised: F6, O2 and B5 are now confirmed to compound*

Current	Missing config, a failed worker start, an unconsumed dead-letter alert and a QA block all log and continue (O1, O2, F6).
Problem	The system's most likely failure mode is *doing nothing while looking healthy*.
Proposed	Introduce a `DEGRADED` concept: a `/readiness` endpoint distinct from `/health` that reports worker-task liveness, DB reachability and each expected-but-absent integration. Declare expected integrations explicitly (e.g. `CMOS_EXPECT_TEAMS=true`) so absence becomes an error rather than a default. Add Azure Monitor alert rules **in Bicep** for dead-letter rate, QA-block rate, budget hard breaches, and loop-completion age.
Benefit	Converts the system's quietest failures into paged ones.
Complexity	Medium.
Dependencies	App Insights (already deployed).

R7 — Make handler retries idempotent

Current	Retries can duplicate Vault rows; acknowledged in-code (B8).
Problem	Duplicate `signals` / `agent_runs` corrupt cost attribution and KPI denominators.
Proposed	Give every handler a deterministic idempotency key — `uuid5(task_id, artefact_role)` — and make Vault creates upsert on it. `get_or_create_campaign` already demonstrates the pattern.
Benefit	Retries become safe; cost figures become trustworthy.
Complexity	Medium — one additive column + unique index per artefact table.
Dependencies	⚠️ `contracts/vault-schema/schema.sql` is frozen. Additive columns/indexes need an explicit freeze decision or a `/v2` namespace.

MEDIUM

R8 — Bound the QA retry loop by cost, not just attempts

Current: ≤10 attempts × 3 completions (B7). **Problem:** one pathological draft can dominate the daily budget; a soft breach only downgrades the tier. **Proposed:** accumulate per-draft spend inside `_run_qa_retry_loop` and exit early past a configured ceiling; add early exit when two consecutive attempts return an identical violation set. **Benefit:** predictable worst-case spend. **Complexity:** low.

R9 — Unify the two brand-safety implementations

Current: prose rules and regex rules for the same six codes (D1). **Problem:** silent drift. **Proposed:** run `safety_suite.py` as a deterministic pre-pass inside `_single_draft_qa_review` and send only the *residual* judgement to the model; the model then handles only genuinely judgement-requiring codes (`unsupported-claim`). **Benefit:** cheaper, faster, more consistent; removes the drift. **Complexity:** medium — needs an eval comparison first. **Note:** `brand_rules.reconcile_violations` is already a partial step in this direction.

R10 — Separate analytics storage from the transactional database

Current: four schemas on one server (B1, SC3). **Proposed:** move `analytics` to its own database or a read replica; point Power BI at that. **Benefit:** removes reporting contention; shrinks blast radius. **Complexity:** medium. **Dependencies:** cross-database reads for `cost_per_accepted_asset` — resolve with a nightly extract rather than a live join.

R11 — Replace runtime `az` shell-outs with configuration

Current: service discovery via subprocess (B4, D5). **Proposed:** thread every service URL in as a Container Apps env var at deploy time (Bicep already knows the FQDNs); keep `resolve_live_fqdn` as a local-dev fallback only. **Benefit:** removes a runtime CLI dependency, a subprocess per cold start, and a duplicated module. **Complexity:** low — Bicep already has the outputs.

R12 — Give `result_ref` a typed contract

Current: untyped JSONB keyed by convention (M3). **Proposed:** a `ResultRef` Pydantic model per producing task type, validated on write in `set_result_ref` and on read in `resolve_lineage_result`. **Benefit:** breaks the silent-rename failure mode. **Complexity:** low-medium.

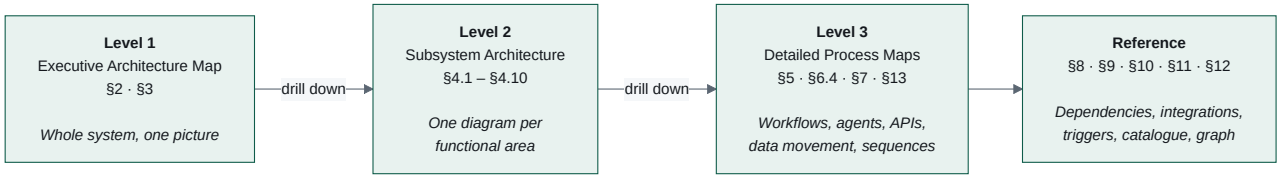
LOW

#	RECOMMENDATION
R13	Delete `functions/task-worker/` (F11). Confirmed not deployed and not built, so deletion is risk-free; the only cost of keeping it is every future reader having to work out that the tier is imaginary.
R14	Remove `la-month-end-reporting-trigger` or ship the `month-end-reporting` loop file (F4).
R15	Get `48-fact-check-verdict/prompt.md` reviewed and drop the "first draft, not approved" banner — or gate the task off until it is (F12).
R16	Move model prices out of `metering.py` into `policy/prices.yaml` alongside the other policy data (M1).
R17	Add a periodic liveness check for `fetch_sources.yaml` URLs — the file's own comment asks for this, noting a retired page "silently narrows" signal quality.
R18	Move Buffer channel/org IDs into environment configuration (SEC4).
R19	Add a reaper for `agent_runs` stuck at `running` after a handler failure (O3).
R20	**Promoted to High.** `mcp-canva` is deployed, holds two live Canva OAuth secrets, and has no caller (F9). Decide deliberately: wire `bulk_create_from_csv` into the carousel handler so function 45's manifest is actually used, or decommission the app and revoke its credentials. Leaving a credentialled, unreachable service running is the one option that has cost and risk but no benefit.

SECTION 16

Three Levels of Visualisation

The diagrams in this document are deliberately layered so a reader can drill down without ever meeting an unreadable diagram.



LEVEL	DIAGRAMS	ANSWERS
1 — Executive	§2 master map, §3 layer stack, §3 trust boundary	*What is this system made of?*
2 — Subsystem	10 subsystem diagrams (§4)	*How does one functional area work?*
3 — Process	5 process flows (§5), agent interaction graph (§6.4), 4 data-flow diagrams (§7), 4 journeys (§13)	*What exactly happens, step by step?*
Reference	Dependency graph (§8), integration map (§9), trigger map (§10), component catalogue (§11), relationship graph (§12)	*What depends on what, and where does it live?*

SECTION 17

Documentation Standards Used

- **Mermaid** for all diagrams: `flowchart` for architecture and data flow, `sequenceDiagram` for journeys, `stateDiagram-v2` for the task lifecycle.
- **Consistent node shapes and edge semantics** — the legend at the top of this document applies to every diagram.
- **Consistent terminology.** A *loop* is a YAML task-graph definition; a *task* is one node in a decomposed loop; a *handler* is the Python function that executes a `task_type`; an *agent* is a function package invoked by a handler; a *function package* is the versioned bundle under `functions/`; the *Vault* is Postgres, never Azure Key Vault (which is always written "Key Vault").
- **Multiple focused diagrams over one dense one.** No diagram exceeds ~35 nodes.
- **Fact/interpretation separation.** 🔍 marks something read from the repository; 💬 marks assessment or recommendation.
- **Explicit uncertainty.** Anything not resolvable from source is labelled **Unknown / Requires Confirmation**.

SECTION 18

Evidence Index

Every significant conclusion traced to its source.

CONCLUSION	EVIDENCE
System purpose, layout, worktree workflow	`README.md`
Queue envelope is metadata-only	`contracts/service-bus/{task-envelope.schema.json,spec.md}`
9 Vault tables, FK chains, append-only convention, join convention	`contracts/vault-schema/schema.sql`
Gate-token claims, alg allowlist, canonical-JSON `resource` packing	`contracts/gate-token/{schema.json,spec.md}`
Completion API shape, runtime schema validation	`contracts/model-gateway/openapi.yaml` , `services/model-gateway/completion.py`
Loop-definition shape, acyclicity is app-level	`contracts/orchestrator/loop-definition.schema.json` , `orchestrator/loop_loader.py`
Contract freezing	`contracts/.frozen-v1.sha256` , `scripts/validate_contracts.py` , `.gitattributes`
6 loop definitions, task graphs, round-34 restructure, proof circuit	`services/orchestrator/loops/*.yaml`
Scanner factory, scoring policy, scan profiles, source candidates	`dispatch.py` `SCANNER_TASKS` ; `functions/_shared/*.yaml`
Approve → publish sweep	`loops/publish-loop.yaml` , `publish_approved_assets_handler`
Measurement join keys	`orchestrator/db.py:551` (utm_campaign_map), `:582` (scheduled_posts)
Worker loop, requeue bound, redelivery reconciliation, event discrimination	`orchestrator/worker.py`
Dispatch table, 18 handlers, 4 exception classes, QA retry loop, lineage walk	`orchestrator/dispatch.py`
Retry/backoff/cascade/idempotency semantics	`orchestrator/state_machine.py`
Task states and transition reasons	`orchestrator/models.py` , `migrations/000{1,3,4}_*.sql`
`result_ref` as data bus	`migrations/0002_task_result_ref.sql` , `orchestrator/db.py`
Advisory-lock ownership	`orchestrator/db.py:try_advisory_lock`
`advance_dependents` all-deps rule	`orchestrator/db.py:advance_dependents`
Deterministic uuid5 decomposition	`orchestrator/decompose.py`
Runtime contracts/functions staging	`orchestrator/config.py` , `.github/workflows/orchestrator-image.yml`
Gateway pipeline order, error hygiene, additive fields	`model-gateway/{completion,main,routing}.py`
Redaction scope + 3 incidents + exemptions	`model-gateway/redaction.py`
Budget downgrade/breach semantics	`model-gateway/{budget.py,policy/budgets.yaml}`
Metering: 3 rows, price table	`model-gateway/metering.py`
Per-process cache, explicit multi-replica scope	`model-gateway/caching.py`
Model routing + live-model validation	`model-gateway/policy/routing.yaml` , `main.py`
Autonomy levels, fail-closed default, RISK-01 removal	`gatekeeper/policy/autonomy.yaml`
Gate-check branches, kill-switch-first ordering	`gatekeeper/app/routers/gate_check.py`
Approval links: single-use, TTL, Easy-Auth approver	`gatekeeper/app/{approval_inbox.py,routers/approval_action.py}`
Adaptive Card / OpenUrl rationale	`gatekeeper/app/teams_client.py`
Kill switch: Postgres not Key Vault, no caching, duplicated	`gatekeeper/app/kill_switch.py` , `tests/test_kill_switch_parity.py`
Publish check ordering and refusal taxonomy	`publisher/app/routers/publish.py`
Verifier standalone, alg pinning, canonical resource	`publisher/app/verifier.py`
Proof-circuit forced dry-run	`publisher/app/{vault_lookup.py,config.py}`
Buffer draft-only invariant	`publisher/app/buffer_client.py` , `mcp/mcp-buffer/{tools.yaml,app/dispatch.py}`
ESP unwired + Mailchimp trade-offs	`publisher/app/esp_client.py`
Vault 9 object types, consent gate, retention, utilisation	`services/vault/vault/*`
Analytics pipeline, idempotent rollups, UTM quarantine, self-validating export	`services/analytics-ingest/analytics_ingest/*`
Fabric export contract	`analytics/contracts/fabric-nightly-export.schema.json`
MCP protocol, fixture mode, allow-list, template lock	`mcp/common/mcp_common/protocol.py` , `mcp/*/tools.yaml` , `mcp/mcp-web/app/tools.py`
Console routes, Easy Auth, interim search	`console/app/{main,auth,routes_reads,routes_write,services}.py`
Telemetry closed enum + length rejection	`services/telemetry-lib/telemetry_lib/attributes.py`
Registry reproducibility, signing, evals, safety suite	`services/registry/{build_registry,signing,eval_harness,safety_suite}.py`
Agent definitions	`functions/*/prompt.md` , `skill.md` , `tools.yaml` , `schema.json` , `evals/`

CONCLUSION	EVIDENCE
Default-deny client naming	`functions/02-brand-steward-qa/permission_check.py` , `docs/permission-register.yaml`
Fetch source allow-list	`functions/09-market-intelligence-director/fetch_sources.yaml`
Fact-check prompt is an unapproved draft	`functions/48-fact-check-verdict/prompt.md`
Trigger schedules and heartbeat bodies	`infra/modules/scheduling/*.bicep`
Nightly job cron and command	`infra/modules/analytics/nightly-ingest-job.bicep`
Infra topology, private endpoints, dependsOn policy	`infra/main.bicep` , `infra/modules/*`
Governance schema tables	`infra/modules/governance/migrations/0001_governance_init.sql`
CI gates, OIDC-only deploys, bundle verification	`.github/workflows/{ci,registry,deploy-infra,orchestrator-image}.yml`
Accepted risks + compensating controls	`docs/accepted-risks.md`
Round-34 batch-gating incident	`docs/content-learnings.md` , `weekly-content-loop.yaml` header
Live e2e verification approach	`tests/e2e/*`

Items labelled Unknown / Requires Confirmation

ITEM	WHY IT COULD NOT BE RESOLVED
Whether Azure Monitor alert rules exist portal-side, outside IaC	Re-confirmed at revision 2 that `infra/` defines none. Portal-created rules cannot be ruled out from the repository — but if they exist, they are invisible to anyone reading the code, which §14.7 O2 treats as a finding in its own right.
Whether `opportunity_cards` is written by anything outside this repo	No writer found in any handler or service.
Current live values of Key Vault secrets (which integrations are actually enabled)	Secrets are correctly absent from the repo; live mode for Buffer/Canva/Teams is credential-gated at runtime.
Whether `mcp-canva` 's Canva OAuth credentials are still live	The app and its Key Vault secret references are deployed; whether the secrets currently hold valid values is not determinable from the repository. It changes how urgent R20 is, so worth checking directly.
Actual production data volumes / current spend against the $5.00 daily loop budget	Requires live telemetry, not source.

Compiled by reverse-engineering the `rooipiet-tech/canvas-marketing-os` repository at commit `5c8ee07` . Diagrams and findings are traceable to the file paths cited throughout. No source code was modified in producing this document.